

BFS Traversal

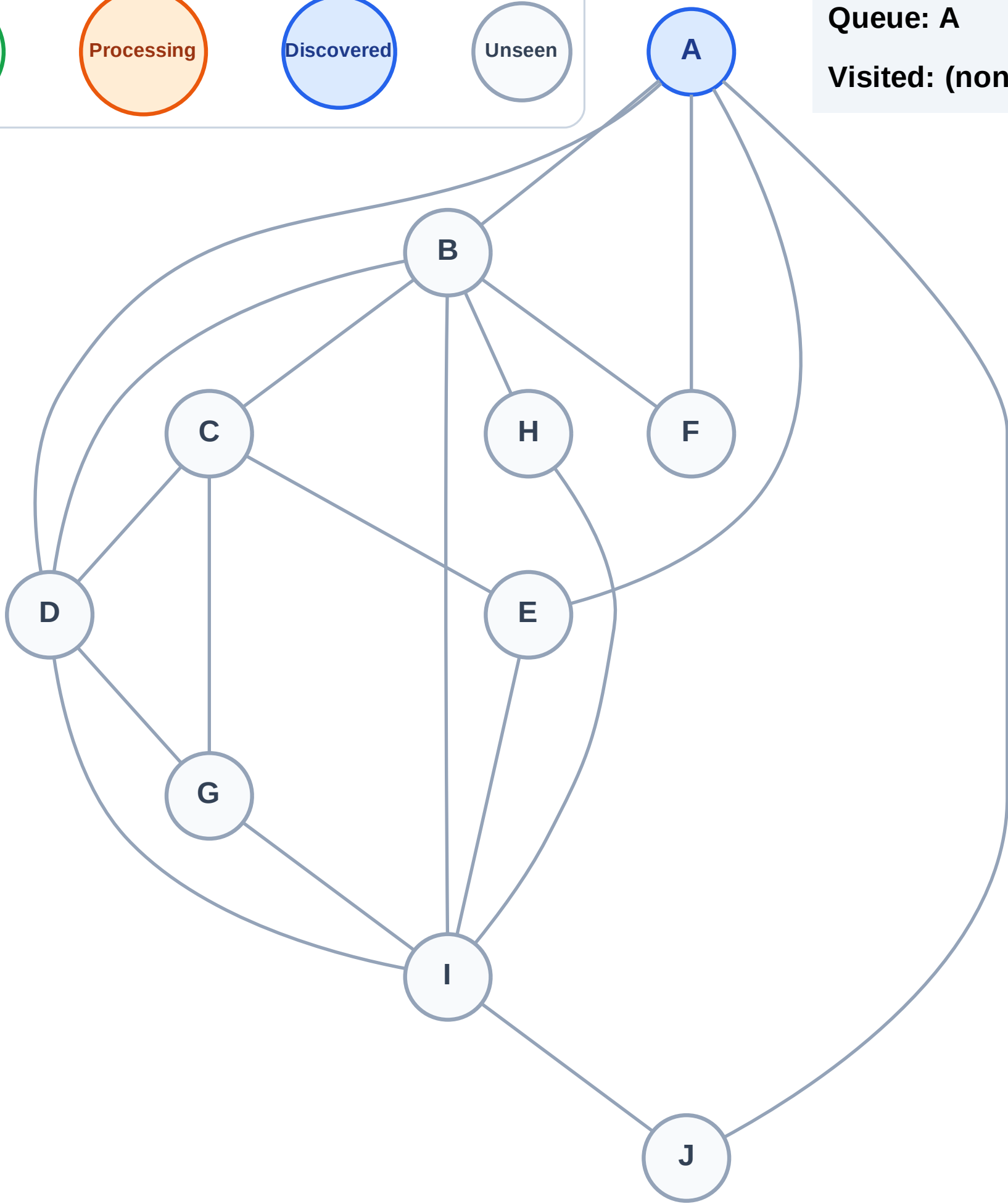
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

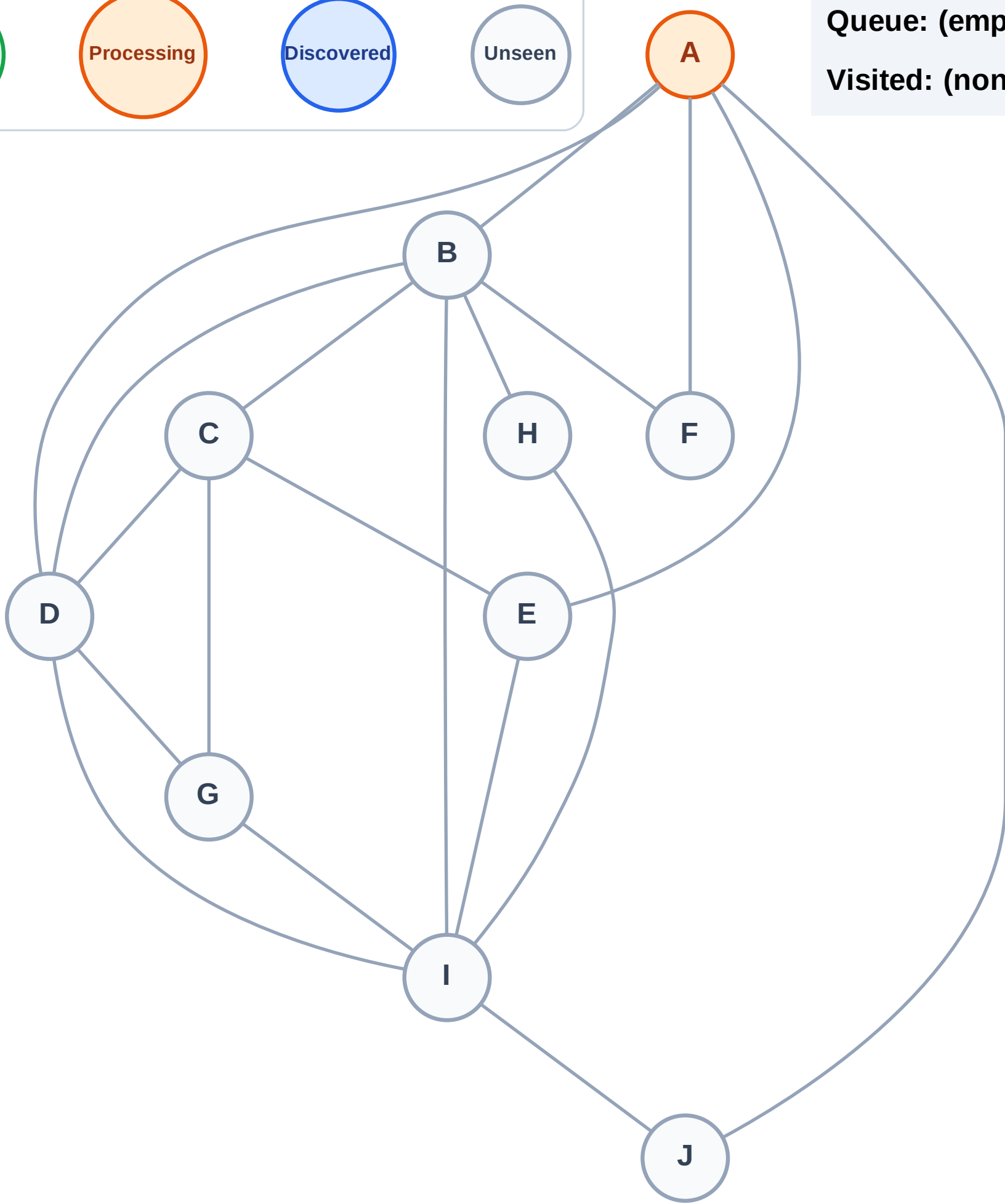
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, D, E, F, J from A

Legend



Complete



Processing

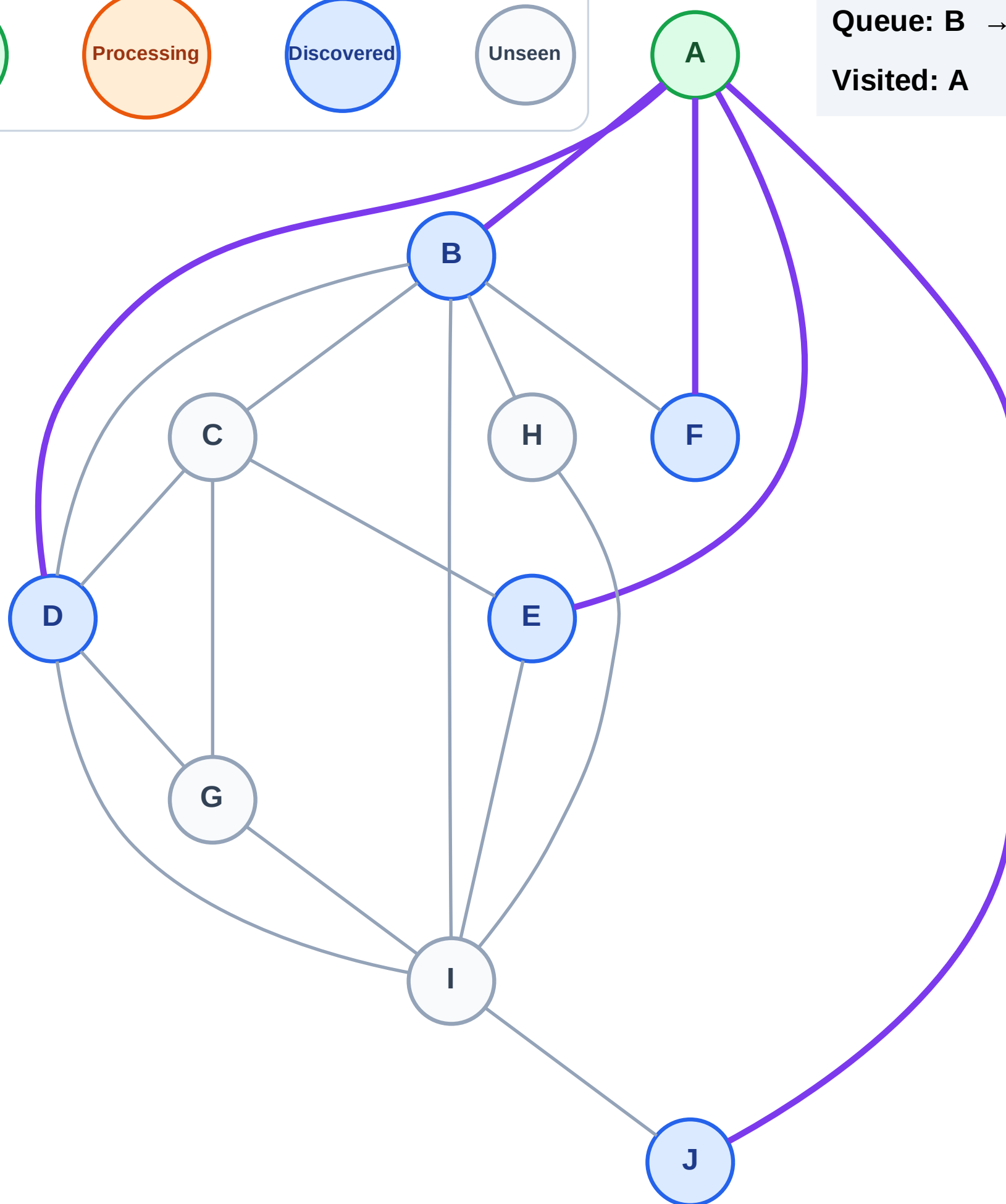


Discovered



Unseen

Queue: B → D → E → F → J
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

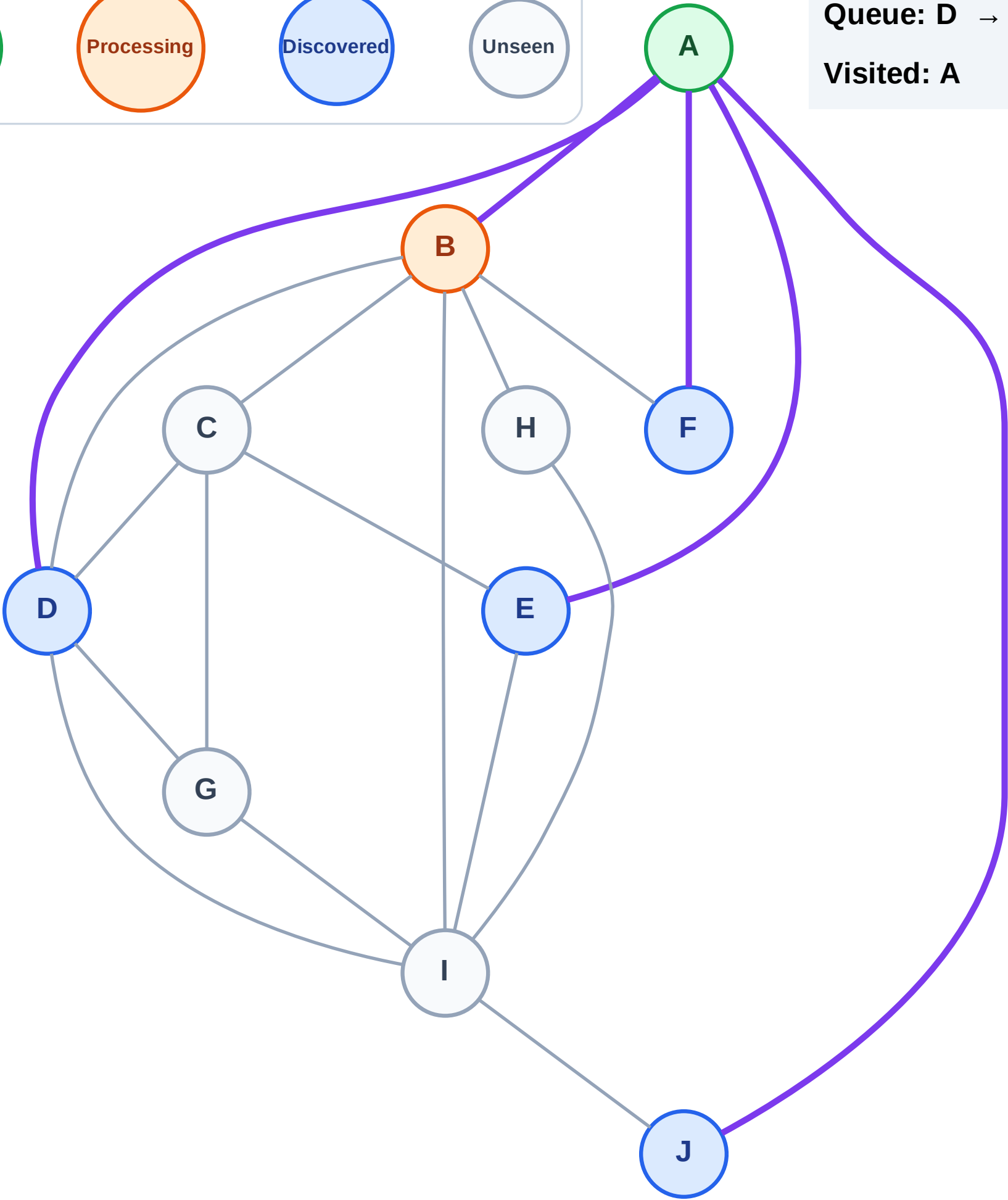
Processing

Discovered

Unseen

Queue: D → E → F → J

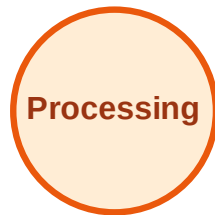
Visited: A



BFS Traversal

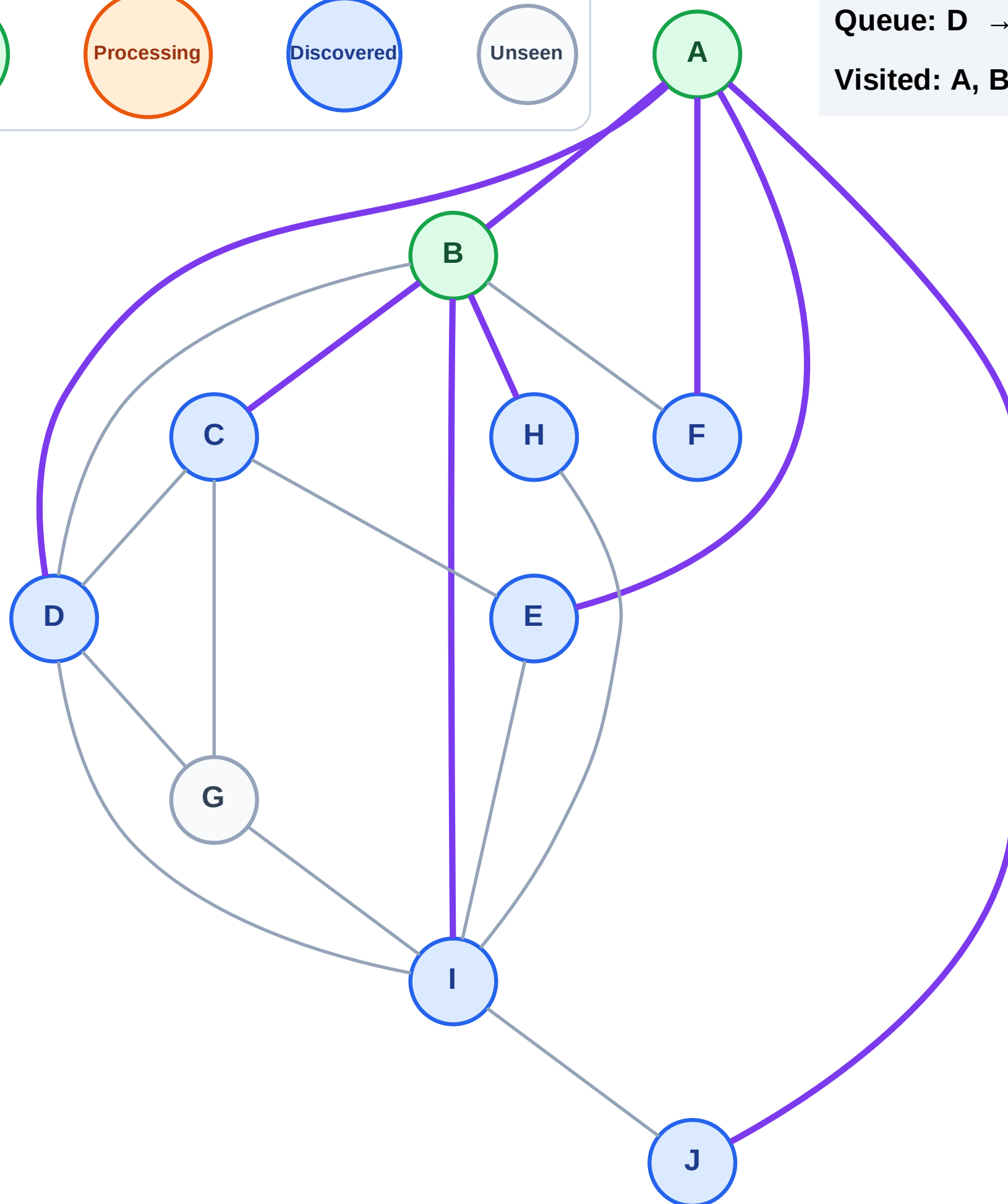
Step 5: Discover C, H, I from B

Legend



Queue: D → E → F → J → C → H → I

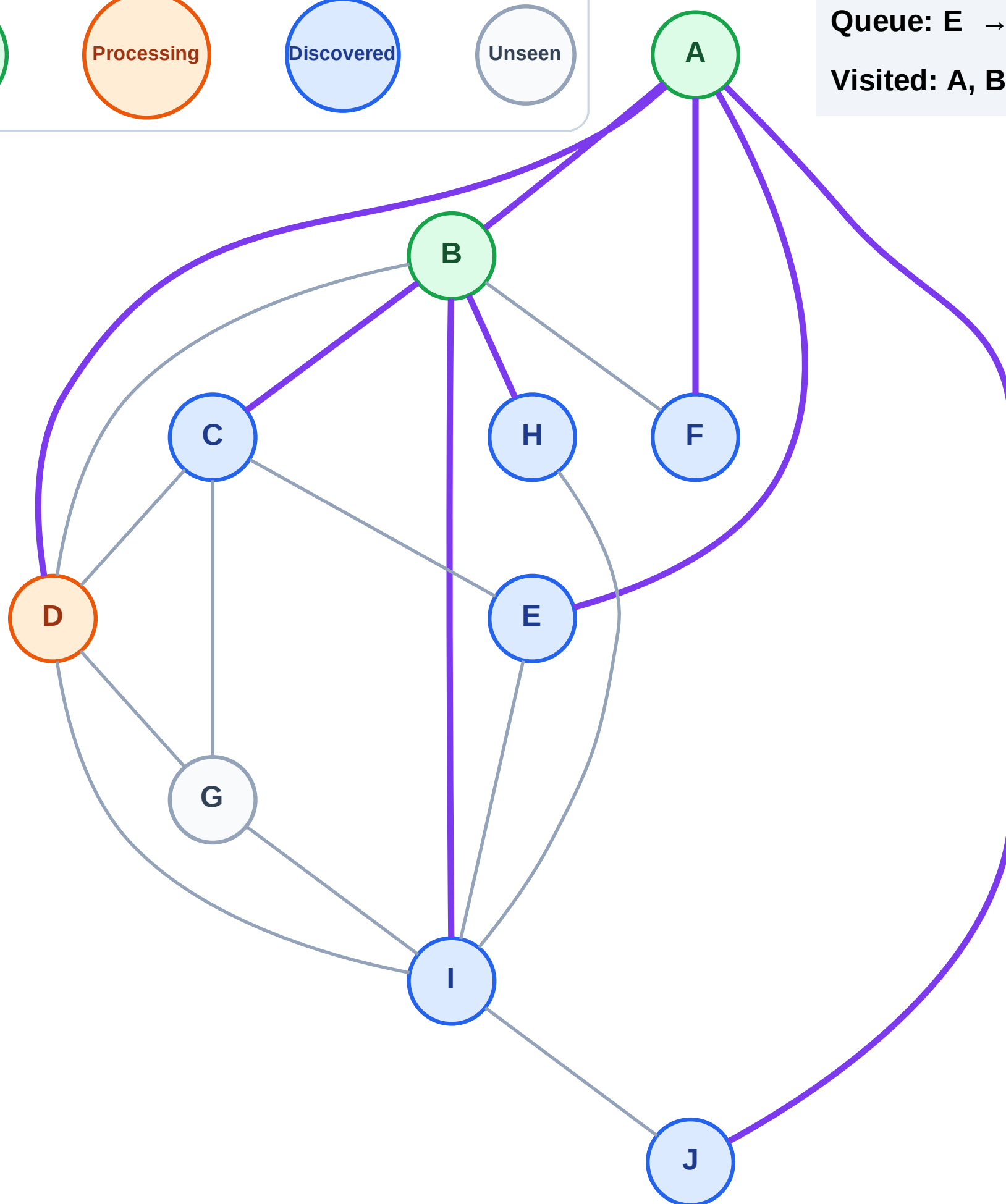
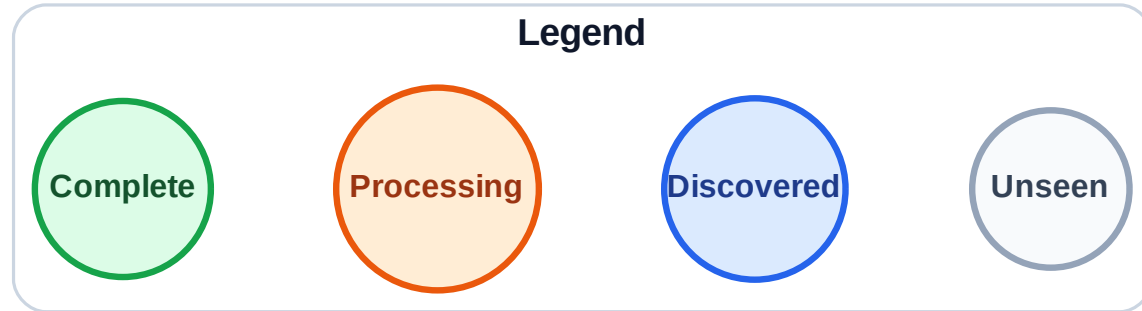
Visited: A, B



BFS Traversal

Step 6: Process node D

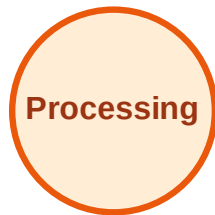
Queue: E → F → J → C → H → I
Visited: A, B



BFS Traversal

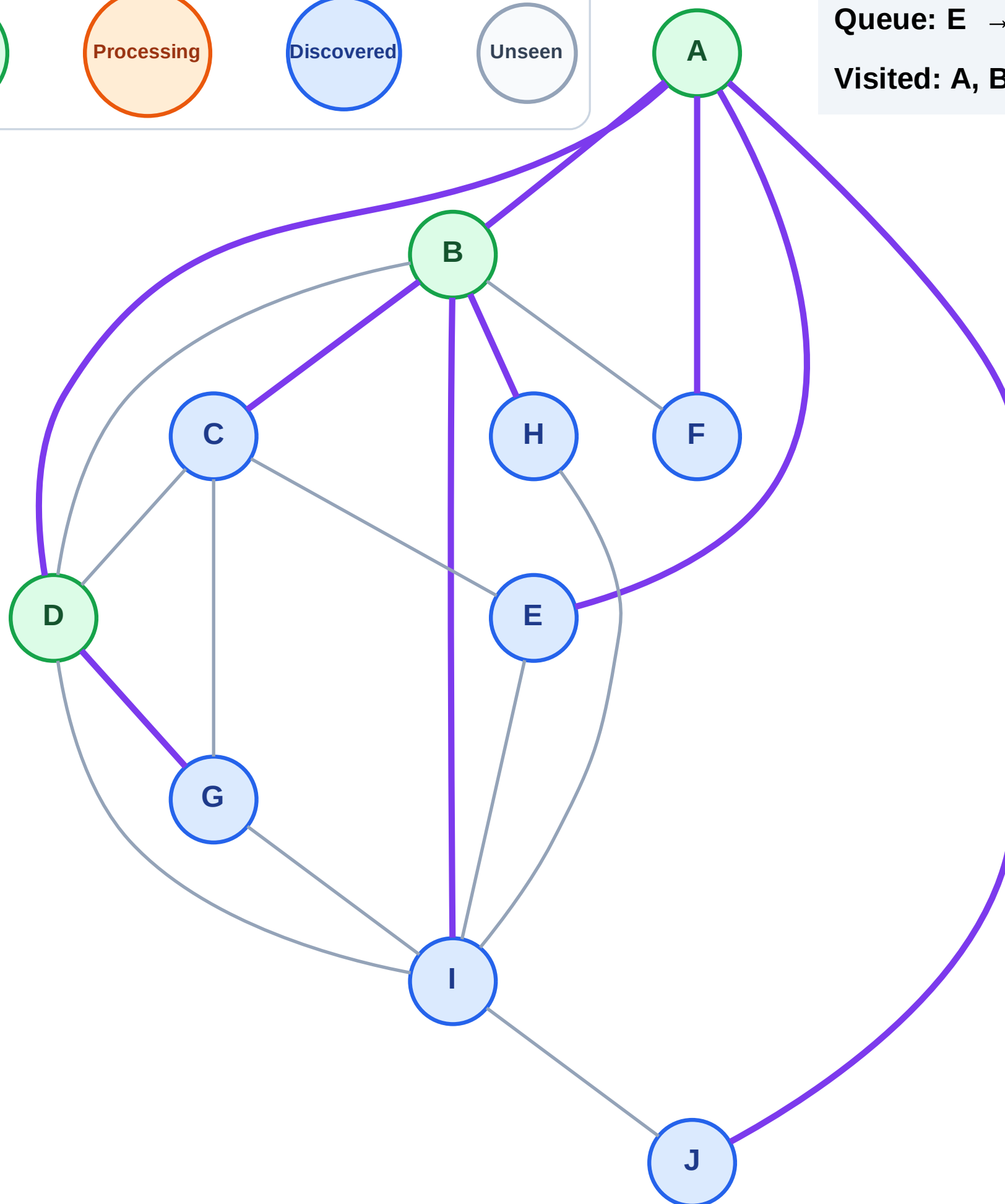
Step 7: Discover G from D

Legend



Queue: E → F → J → C → H → I → G

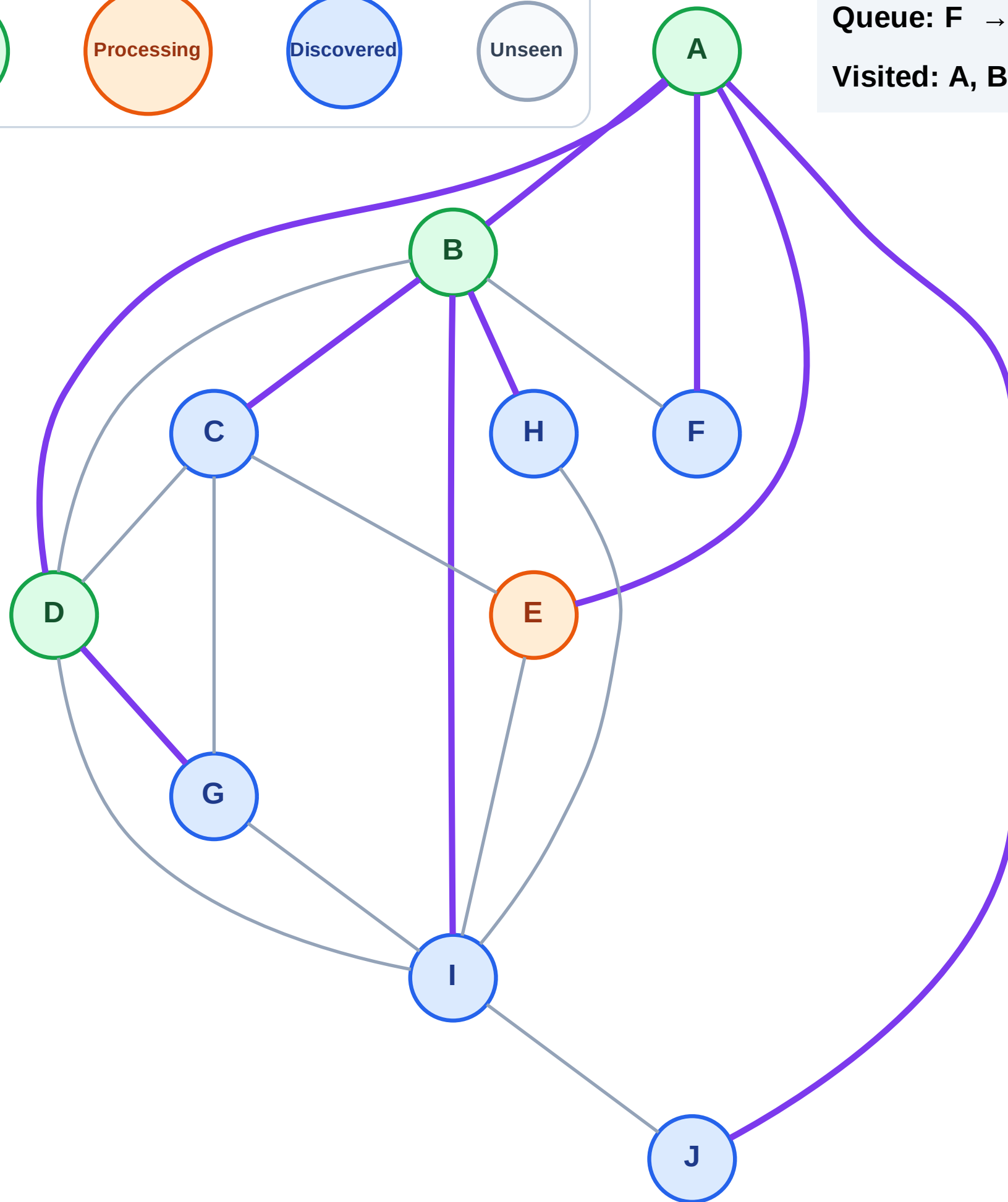
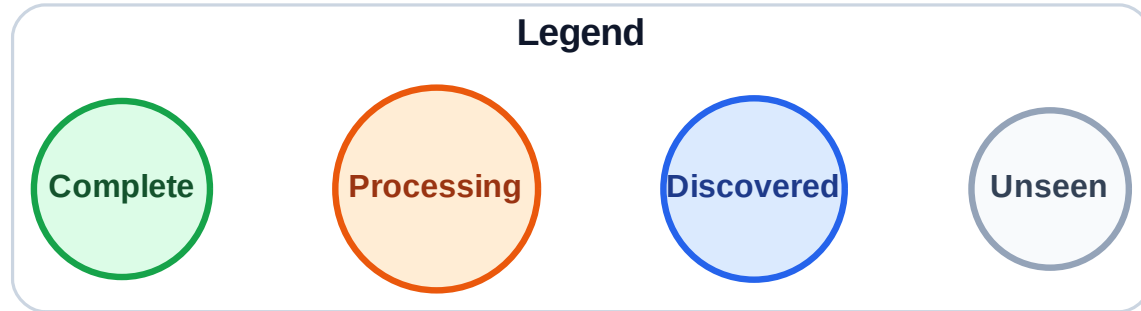
Visited: A, B, D



BFS Traversal

Step 8: Process node E

Queue: F → J → C → H → I → G
Visited: A, B, D



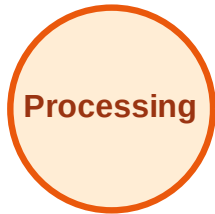
BFS Traversal

Step 9: No new neighbors from E

Legend



Complete



Processing



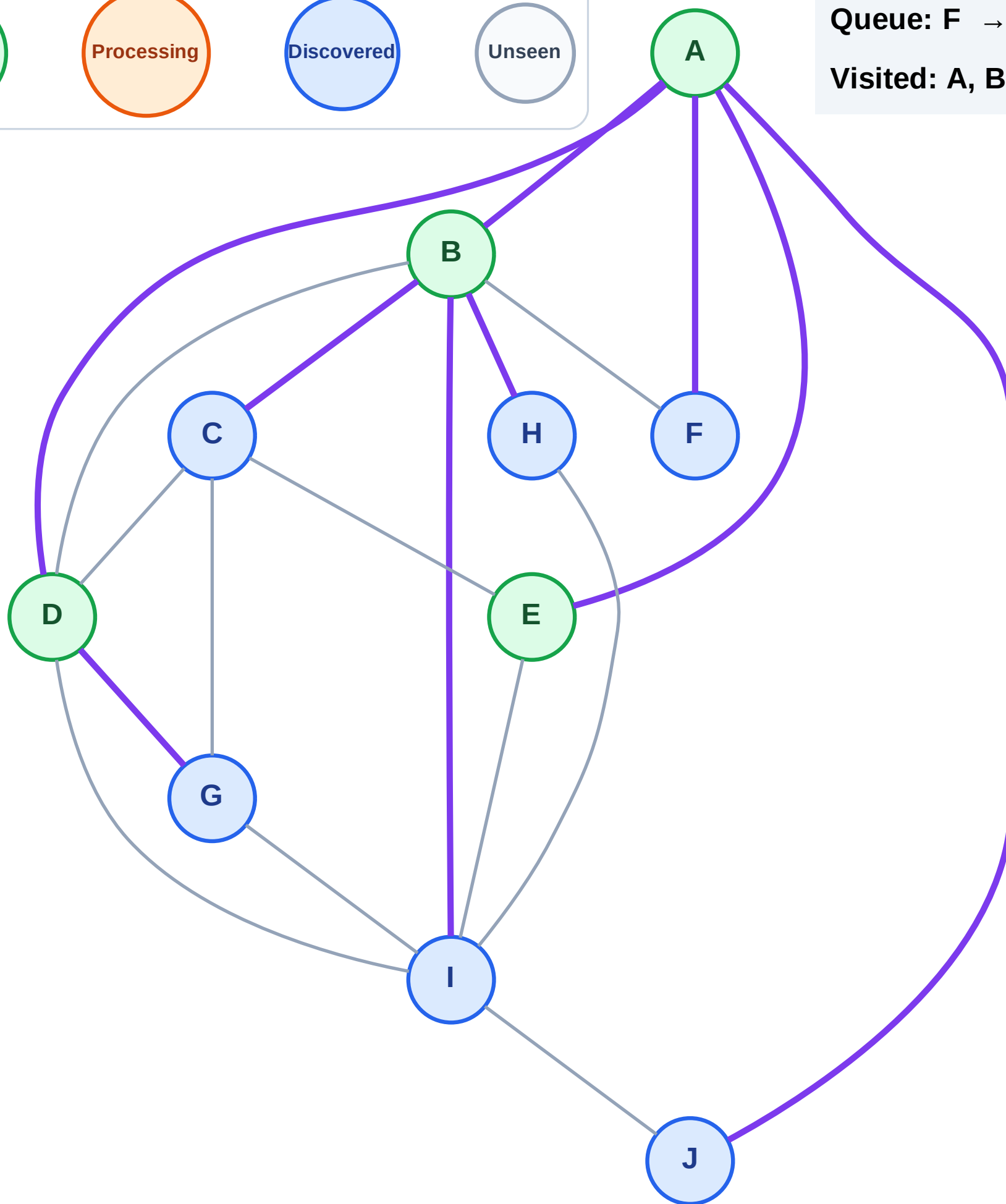
Discovered



Unseen

Queue: F → J → C → H → I → G

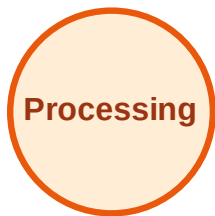
Visited: A, B, D, E



BFS Traversal

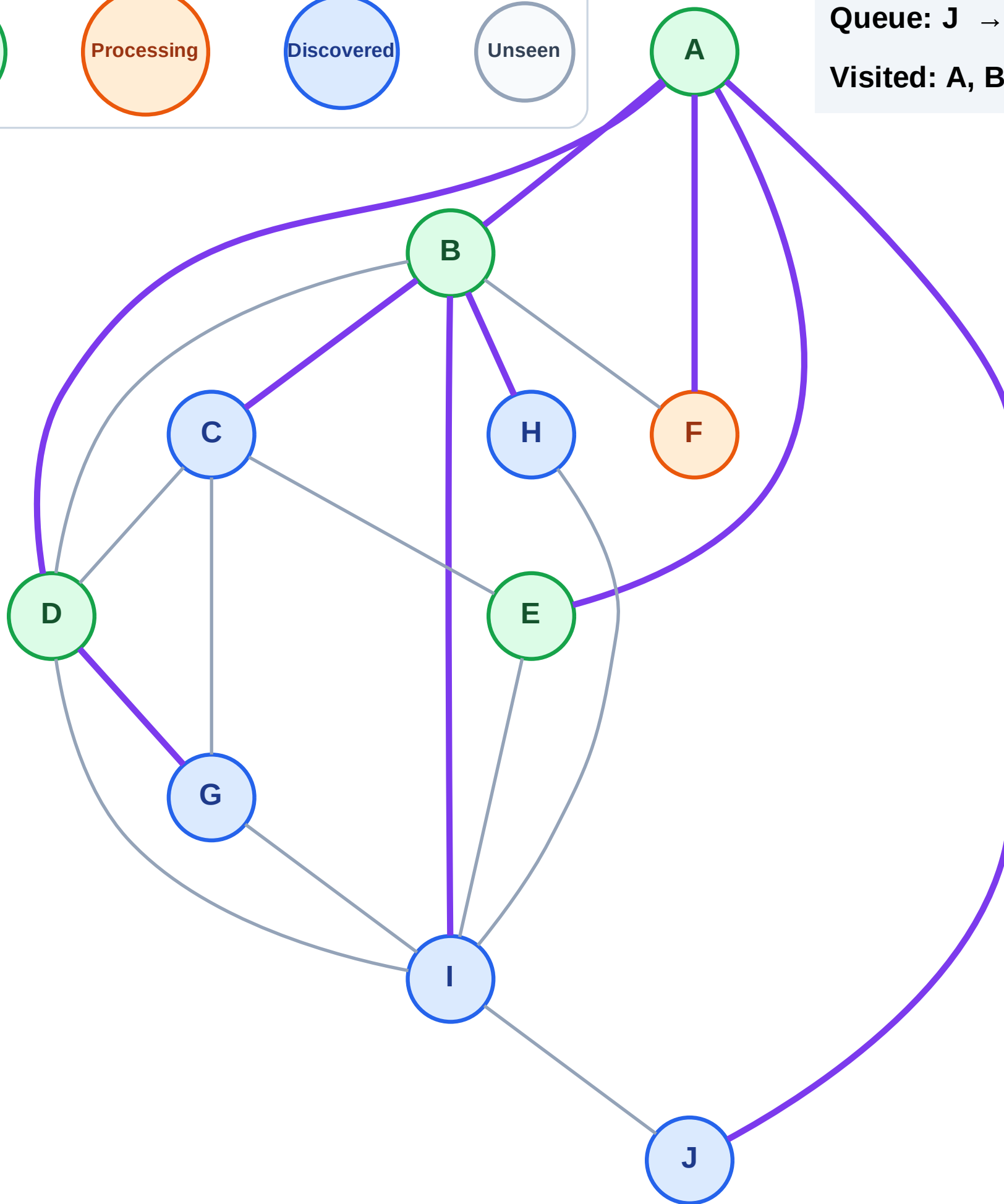
Step 10: Process node F

Legend



Queue: J → C → H → I → G

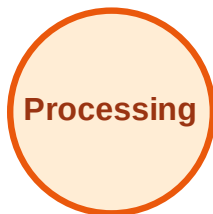
Visited: A, B, D, E



BFS Traversal

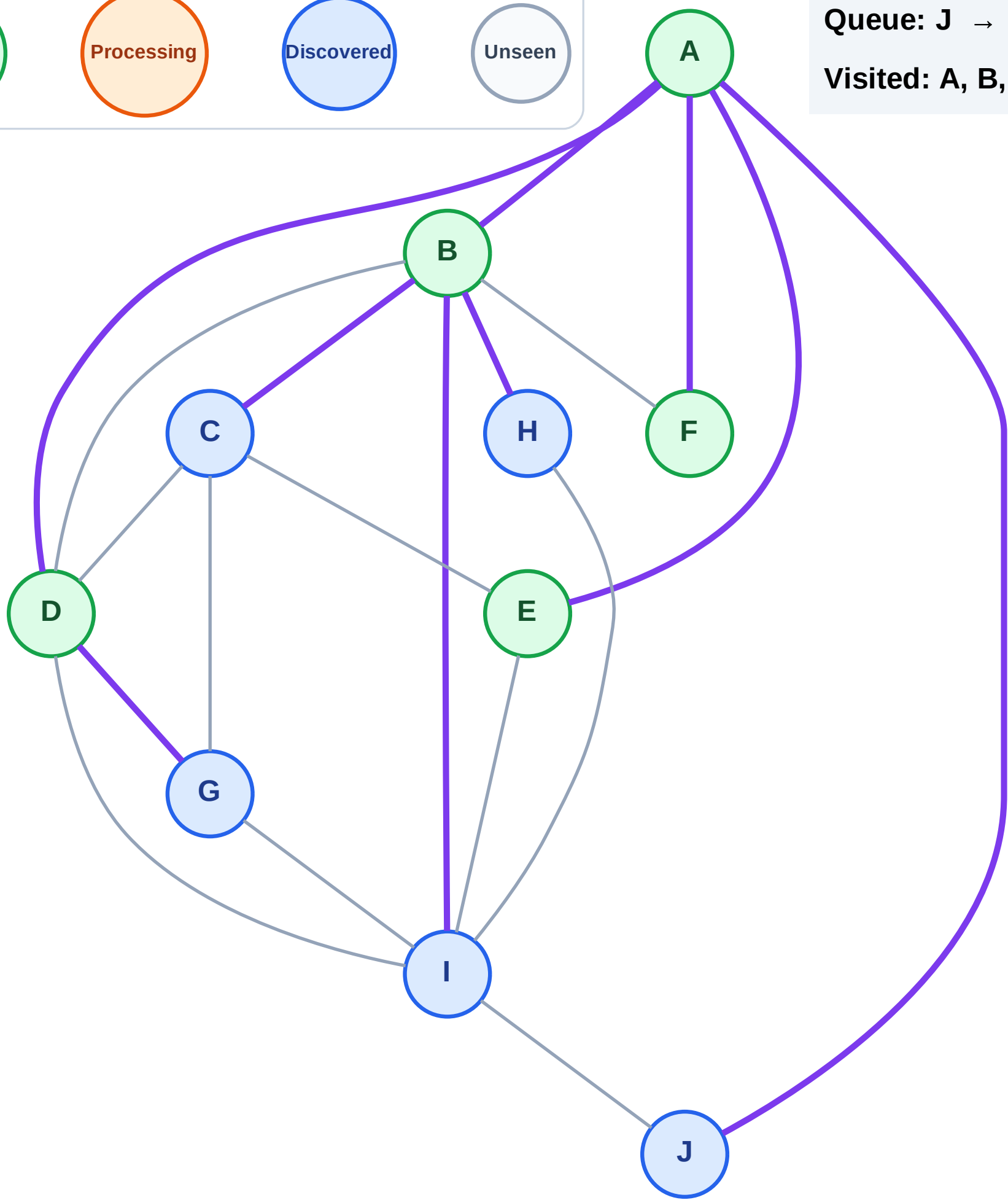
Step 11: No new neighbors from F

Legend



Queue: J → C → H → I → G

Visited: A, B, D, E, F



BFS Traversal

Step 12: Process node J

Legend

Complete

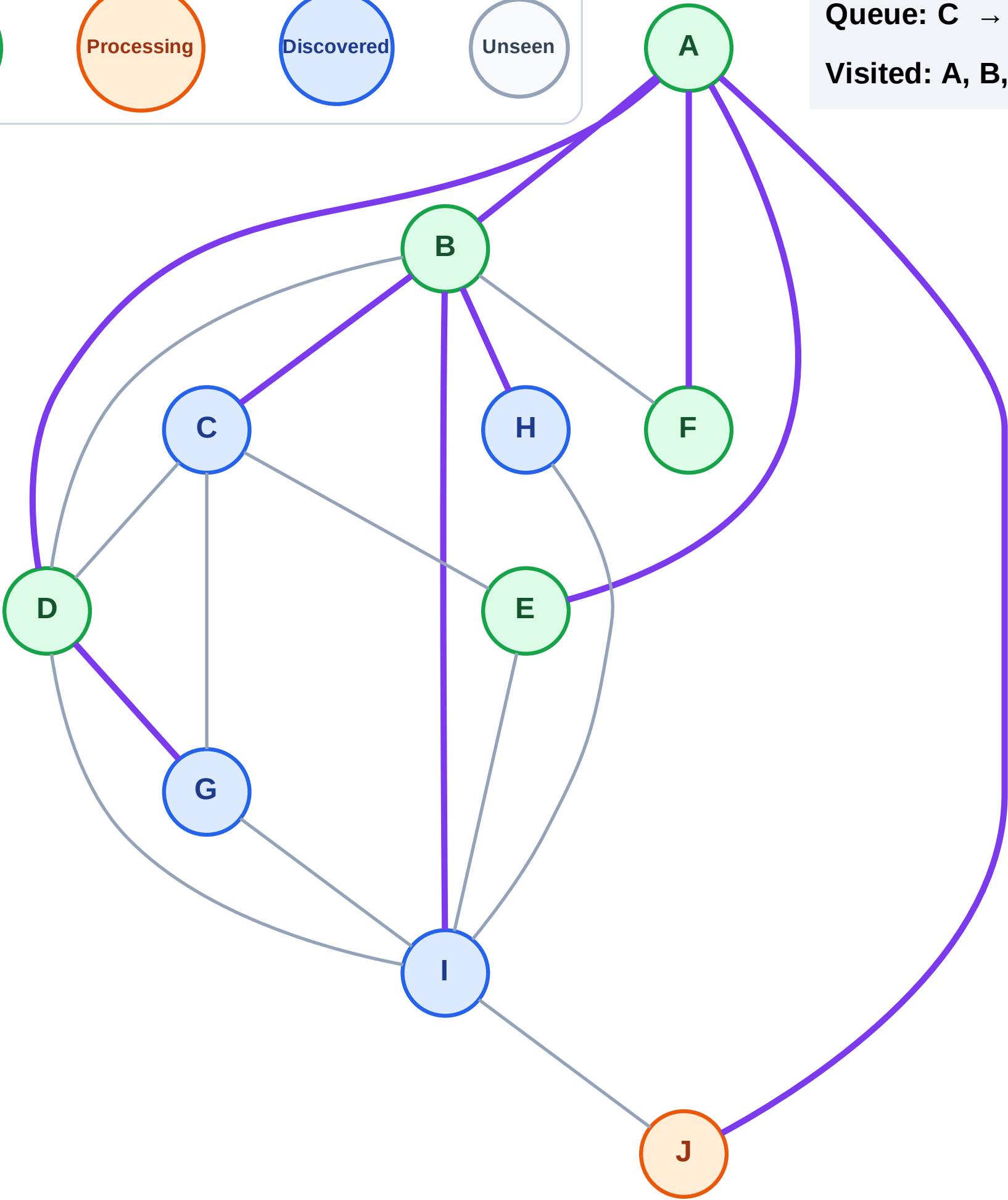
Processing

Discovered

Unseen

Queue: C → H → I → G

Visited: A, B, D, E, F



BFS Traversal

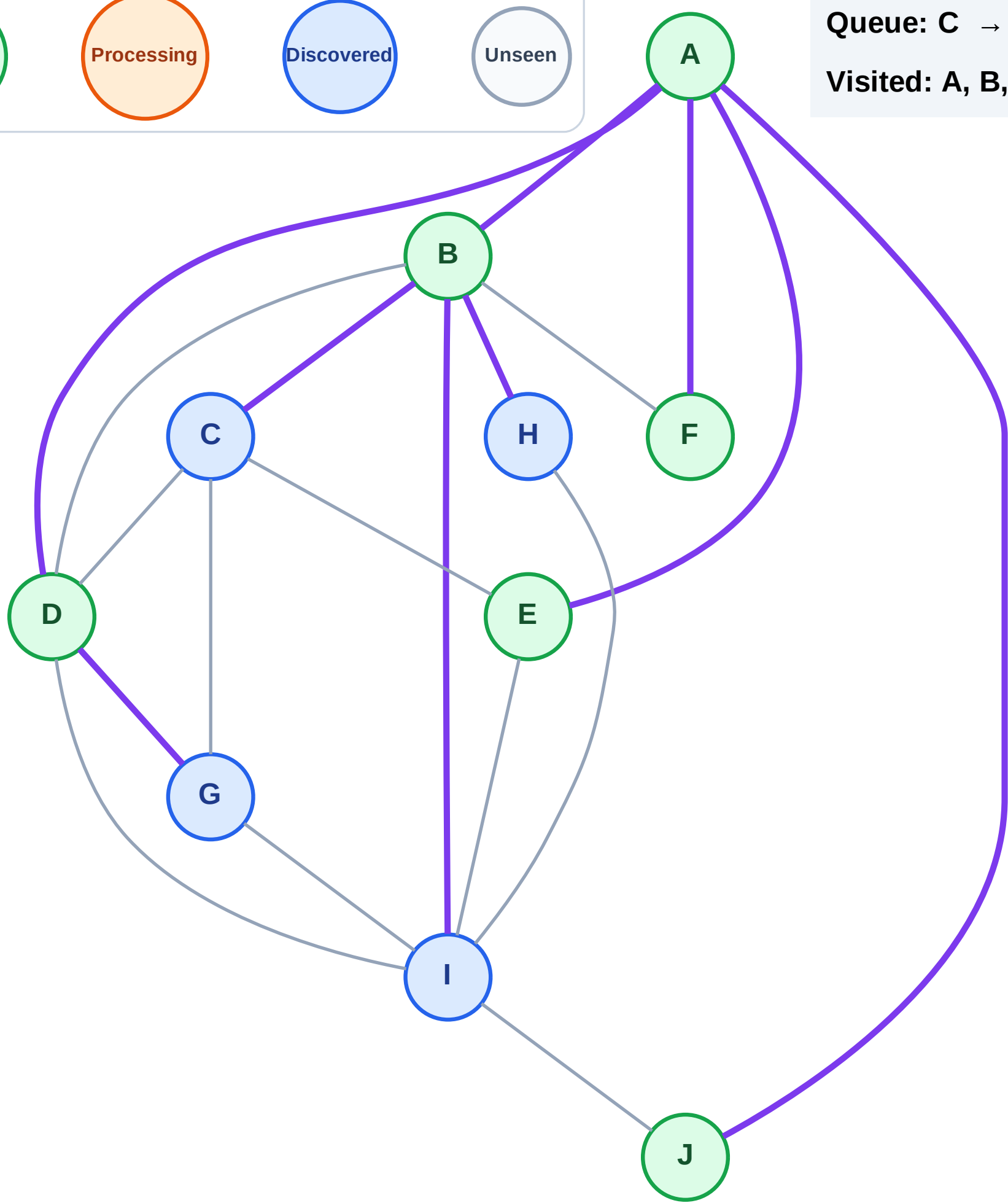
Step 13: No new neighbors from J

Legend



Queue: C → H → I → G

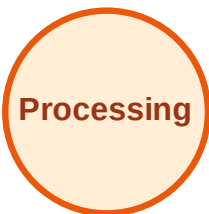
Visited: A, B, D, E, F, J



BFS Traversal

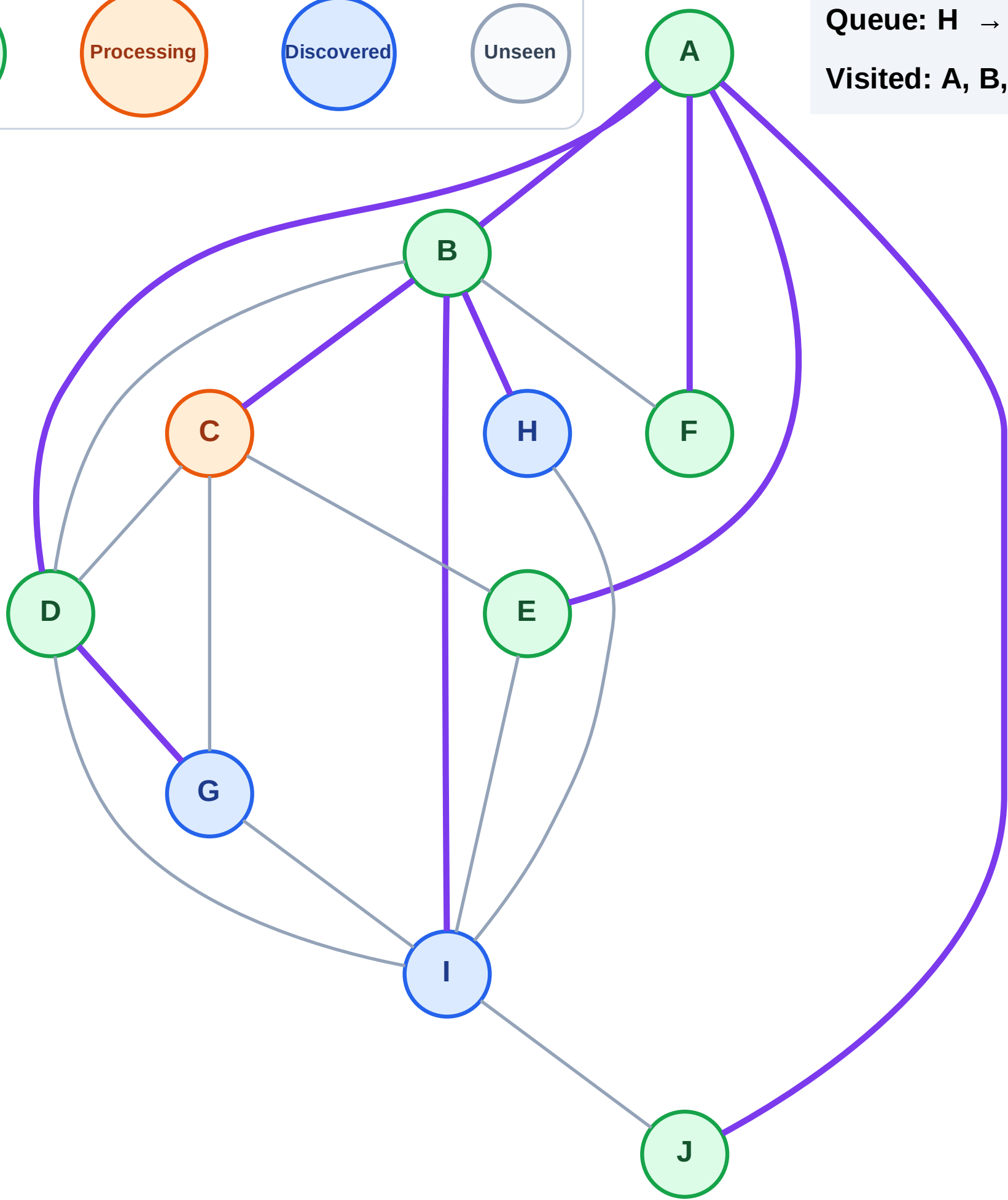
Step 14: Process node C

Legend



Queue: H → I → G

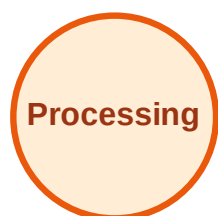
Visited: A, B, D, E, F, J



BFS Traversal

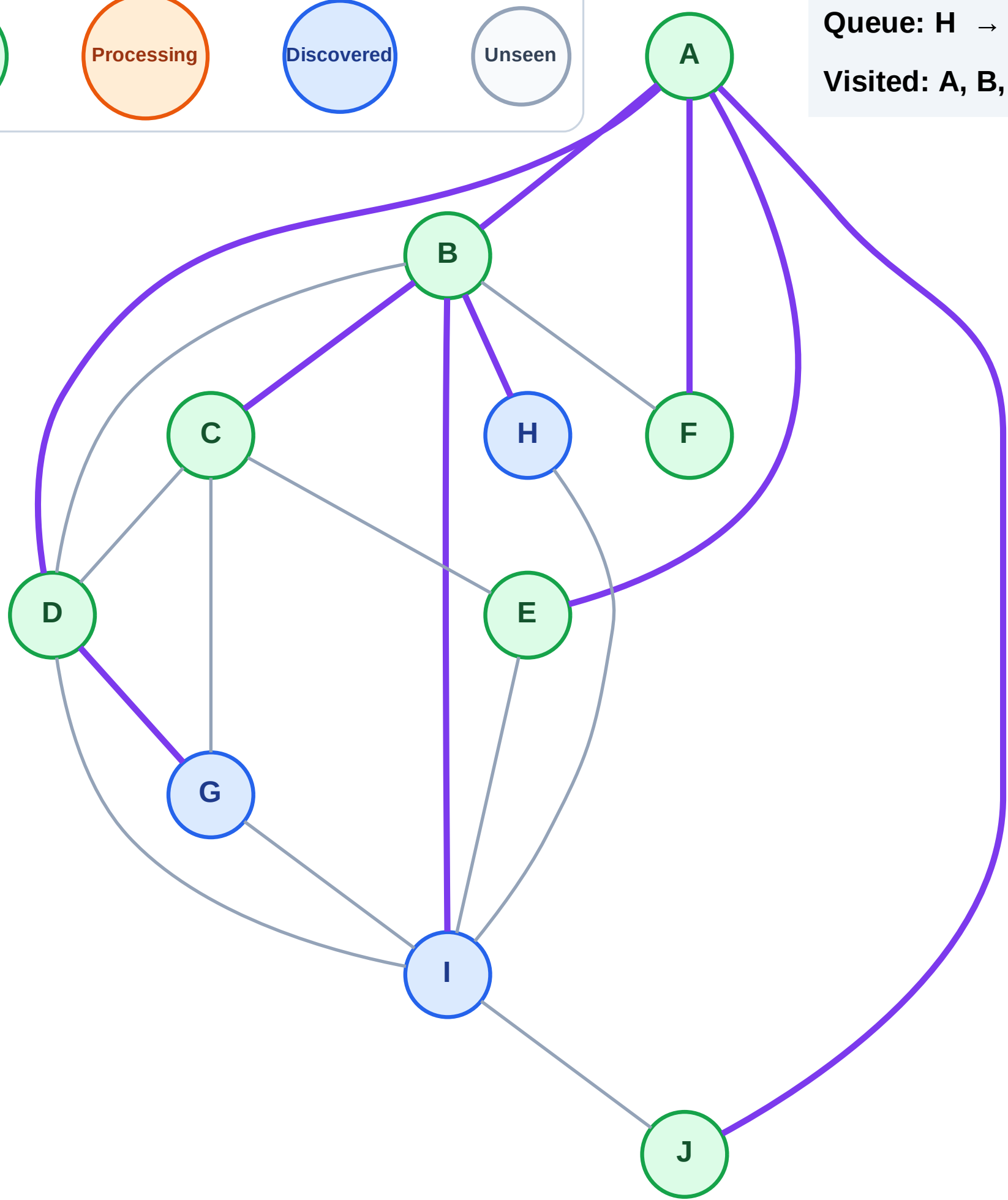
Step 15: No new neighbors from C

Legend



Queue: H → I → G

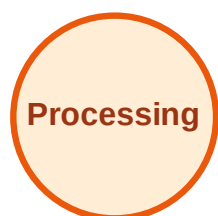
Visited: A, B, C, D, E, F, J



BFS Traversal

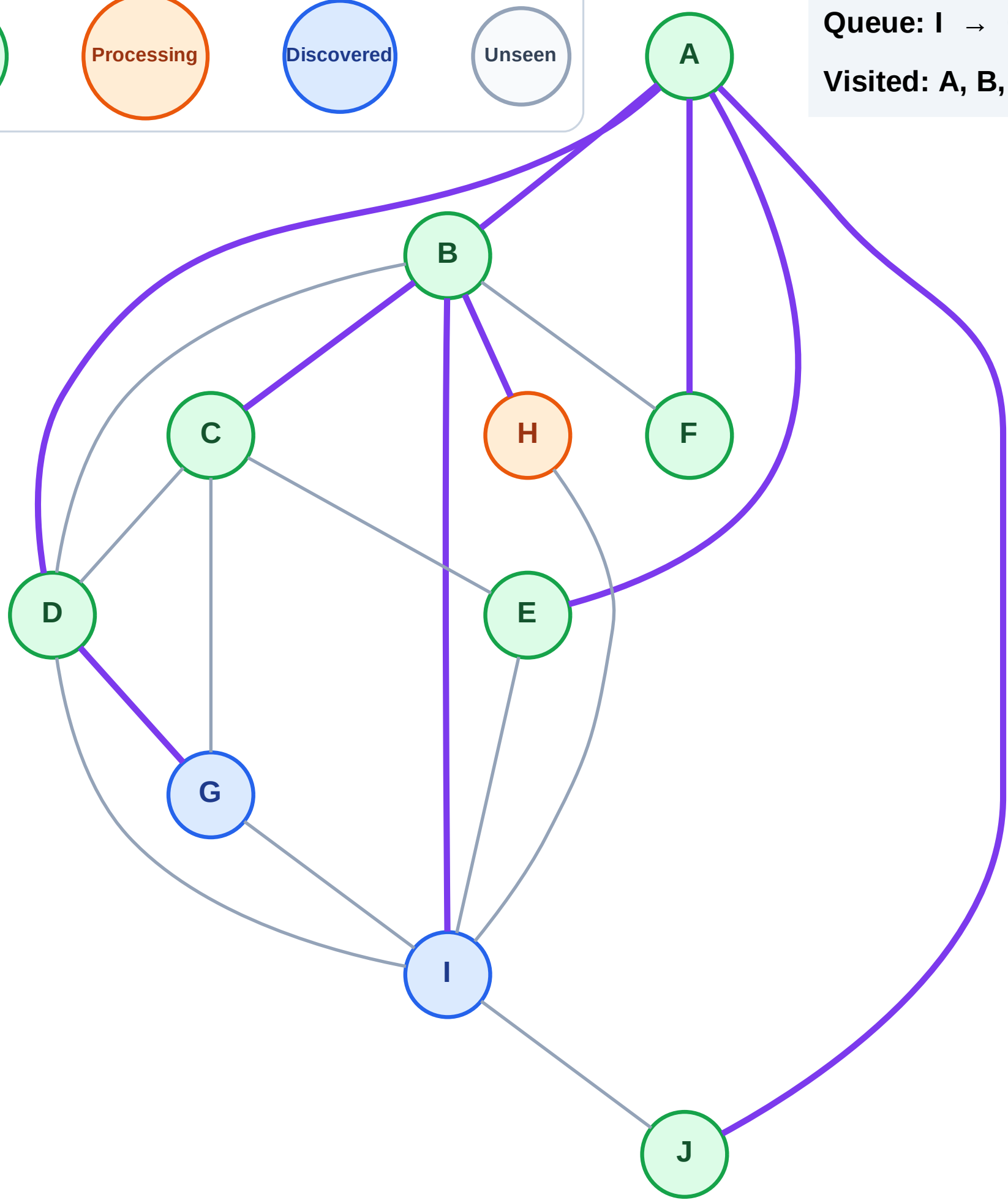
Step 16: Process node H

Legend



Queue: I → G

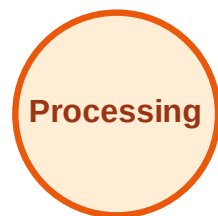
Visited: A, B, C, D, E, F, J



BFS Traversal

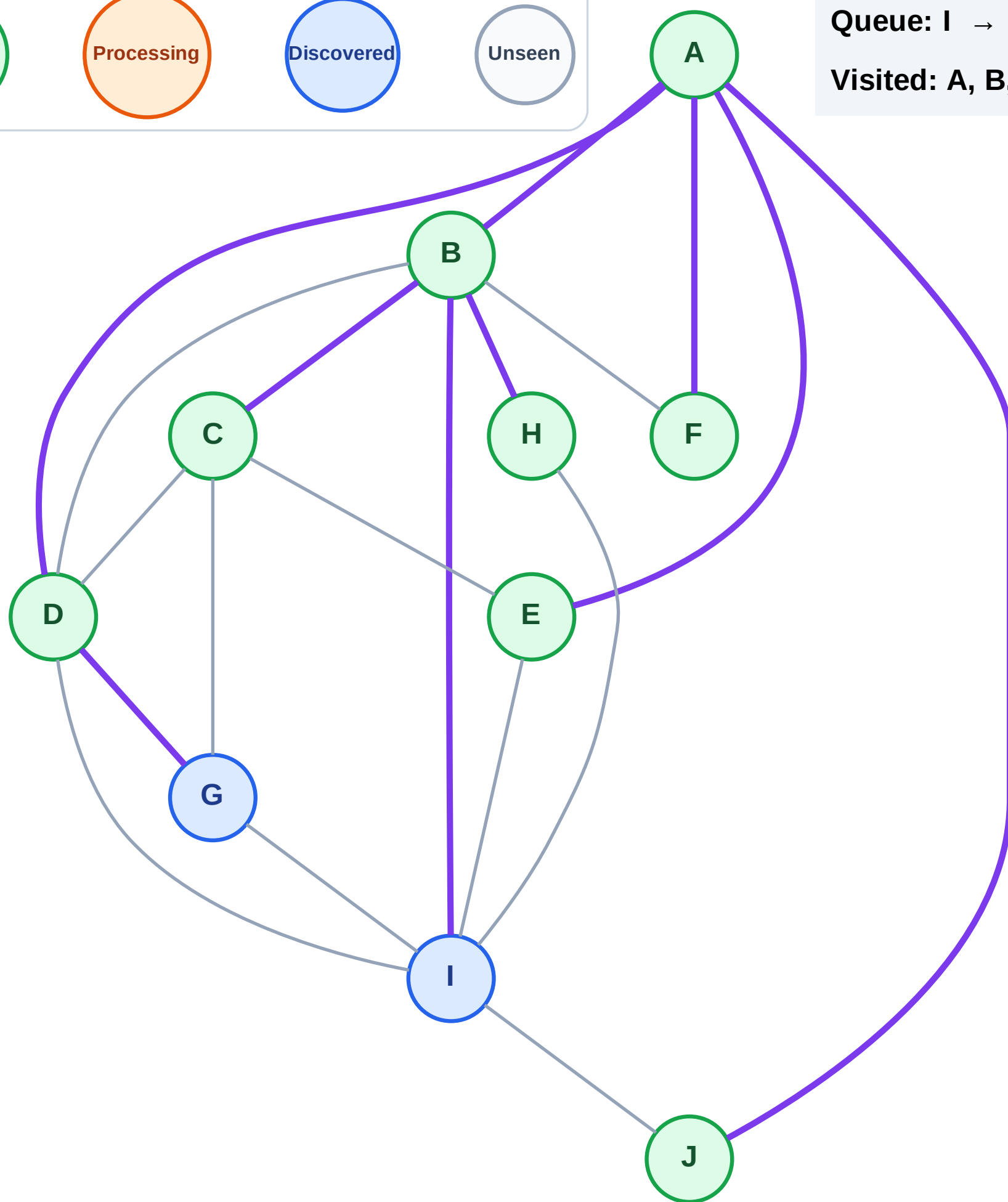
Step 17: No new neighbors from H

Legend



Queue: I → G

Visited: A, B, C, D, E, F, H, J



BFS Traversal

Step 18: Process node I

Legend

Complete

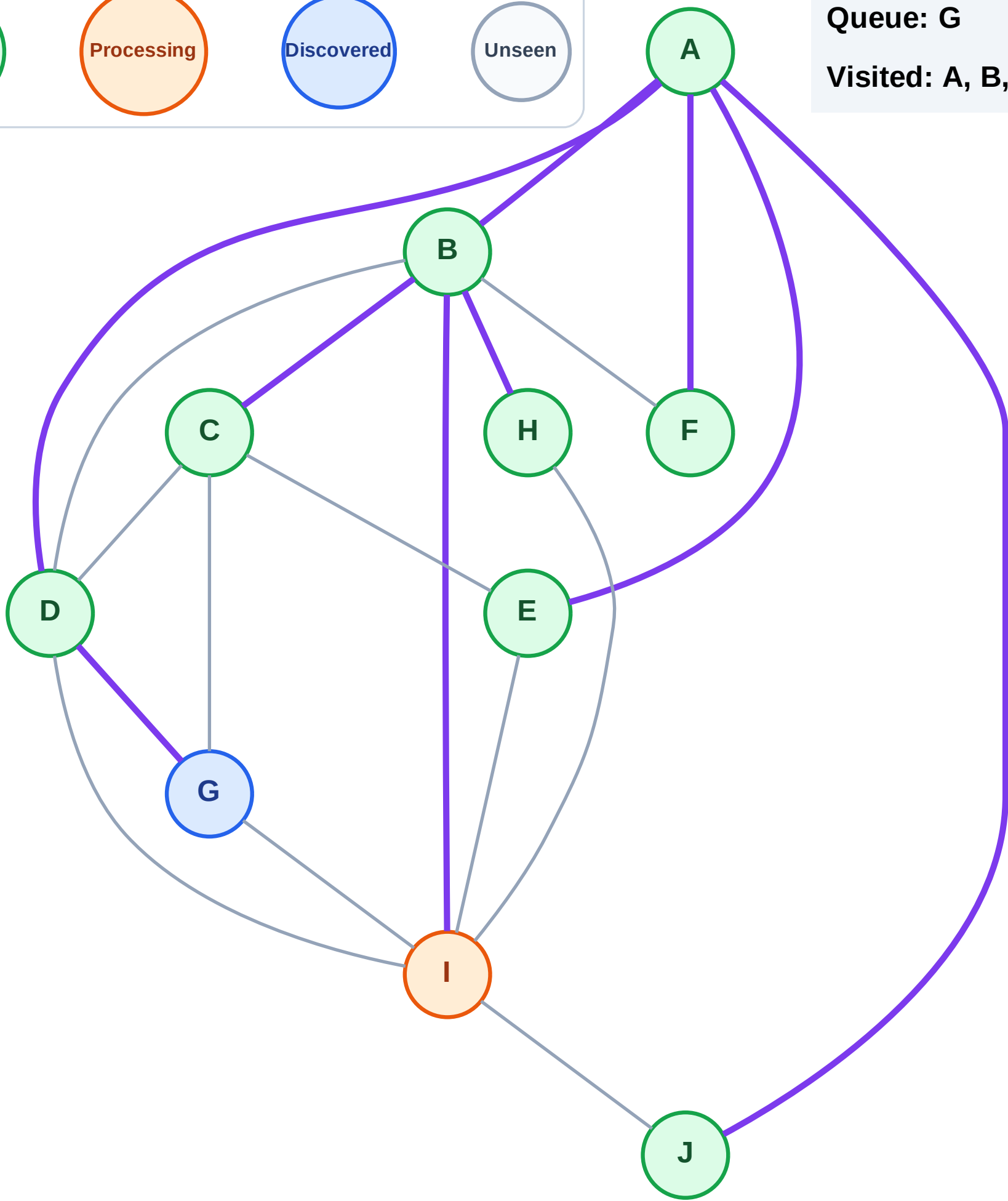
Processing

Discovered

Unseen

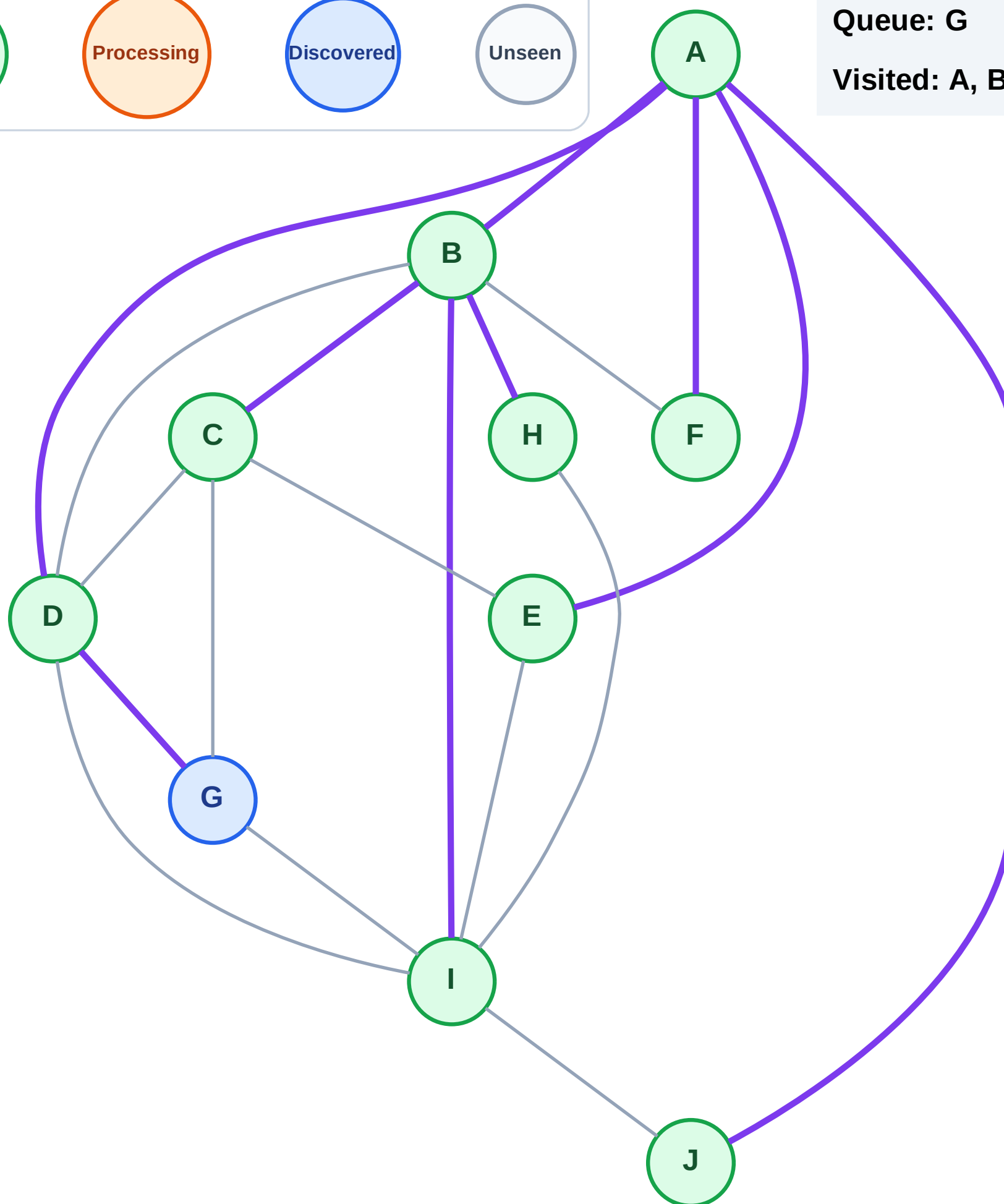
Queue: G

Visited: A, B, C, D, E, F, H, J



Step 19: No new neighbors from I

Step 19: No new neighbors from I



BFS Traversal

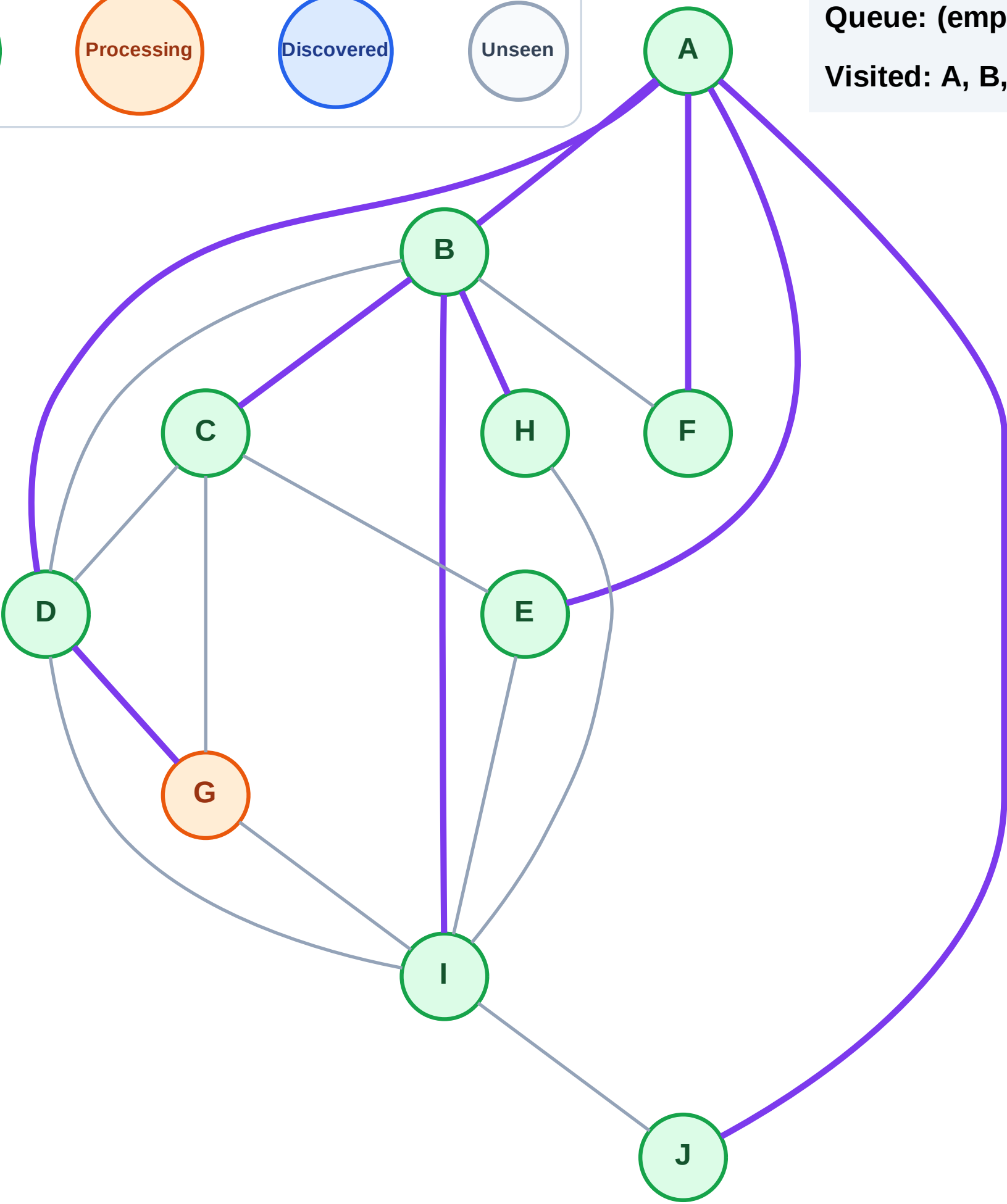
Step 20: Process node G

Legend



Queue: (empty)

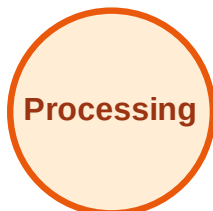
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

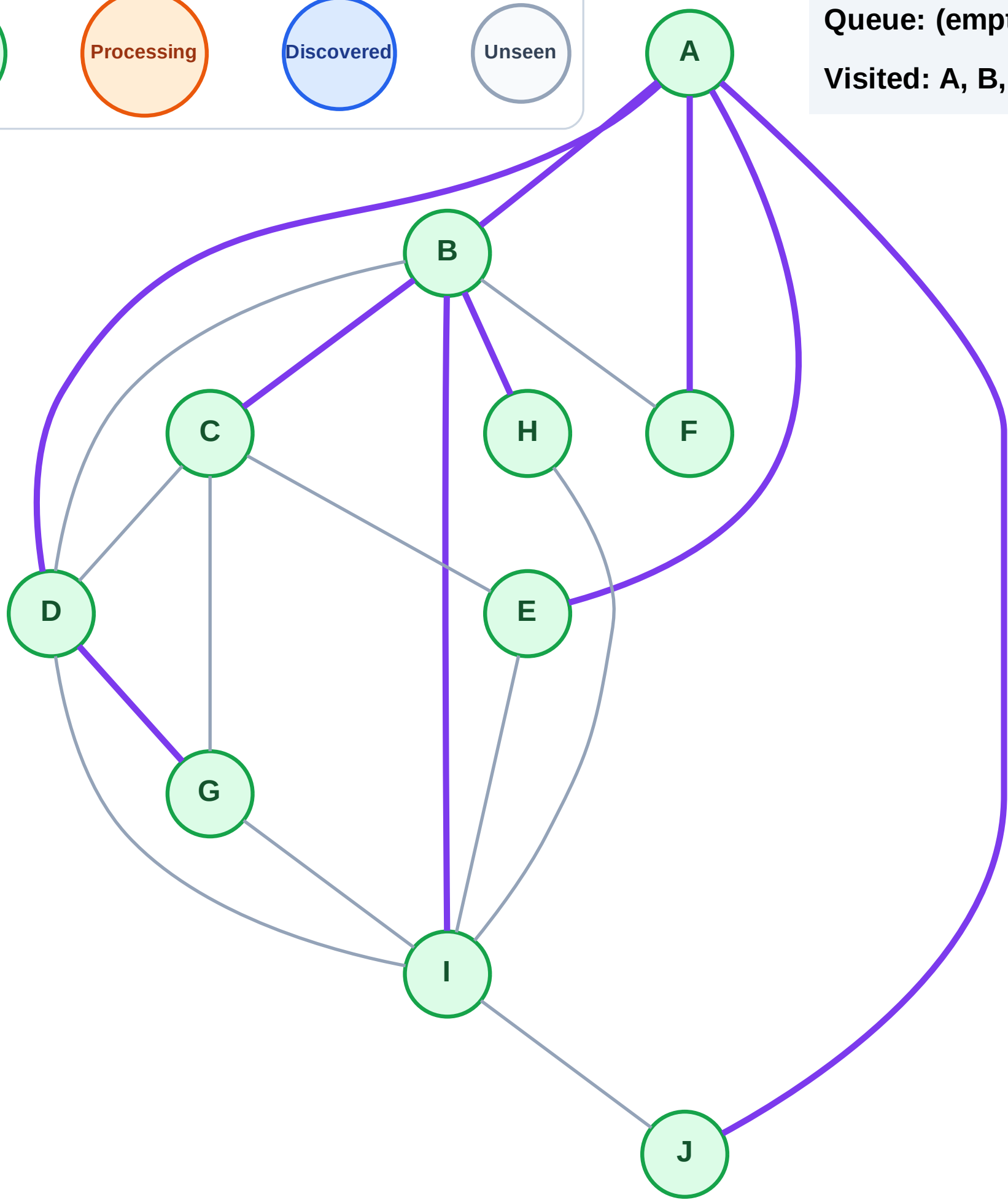
Step 21: No new neighbors from G

Legend



Queue: (empty)

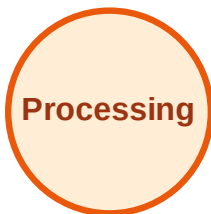
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

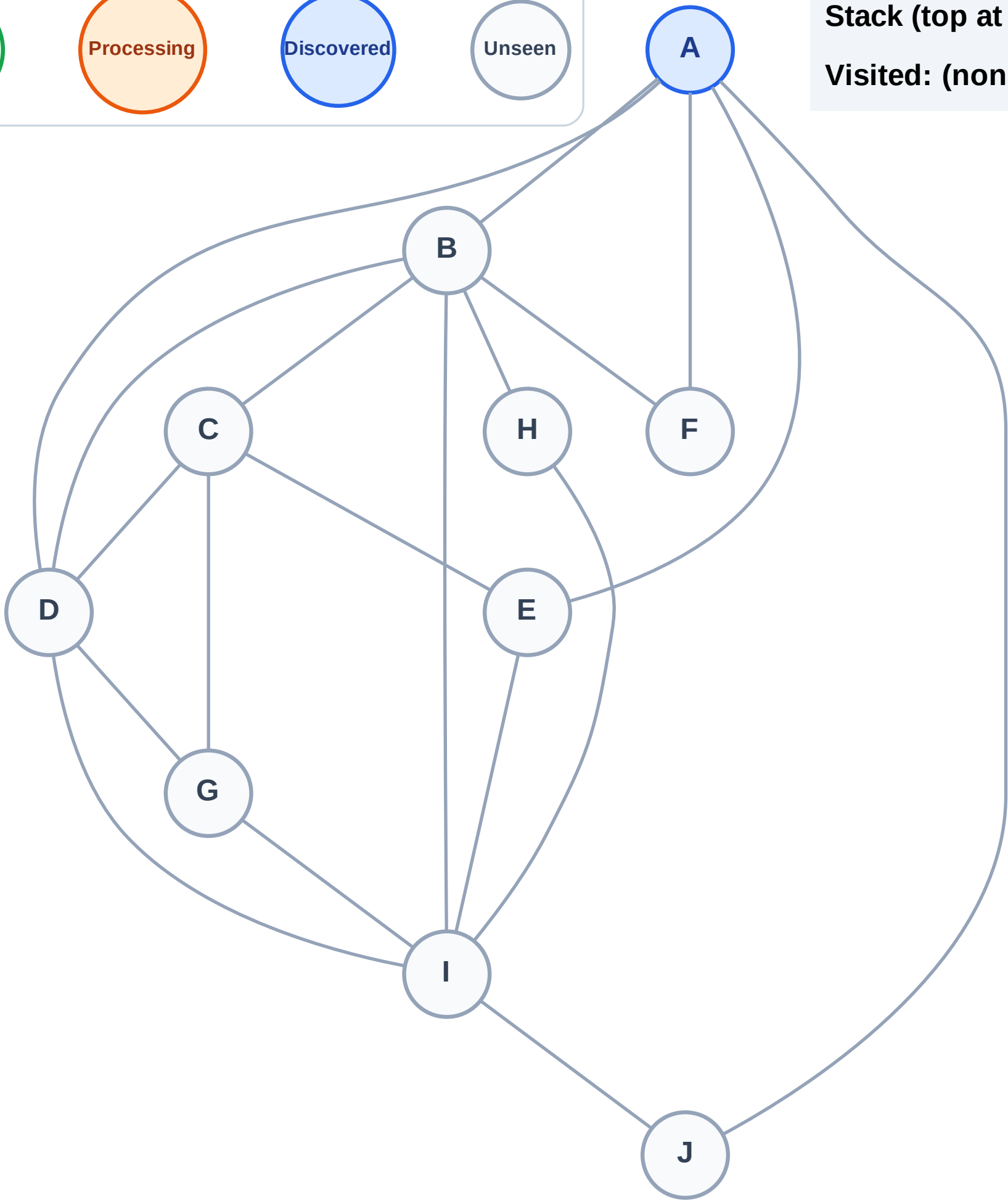
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

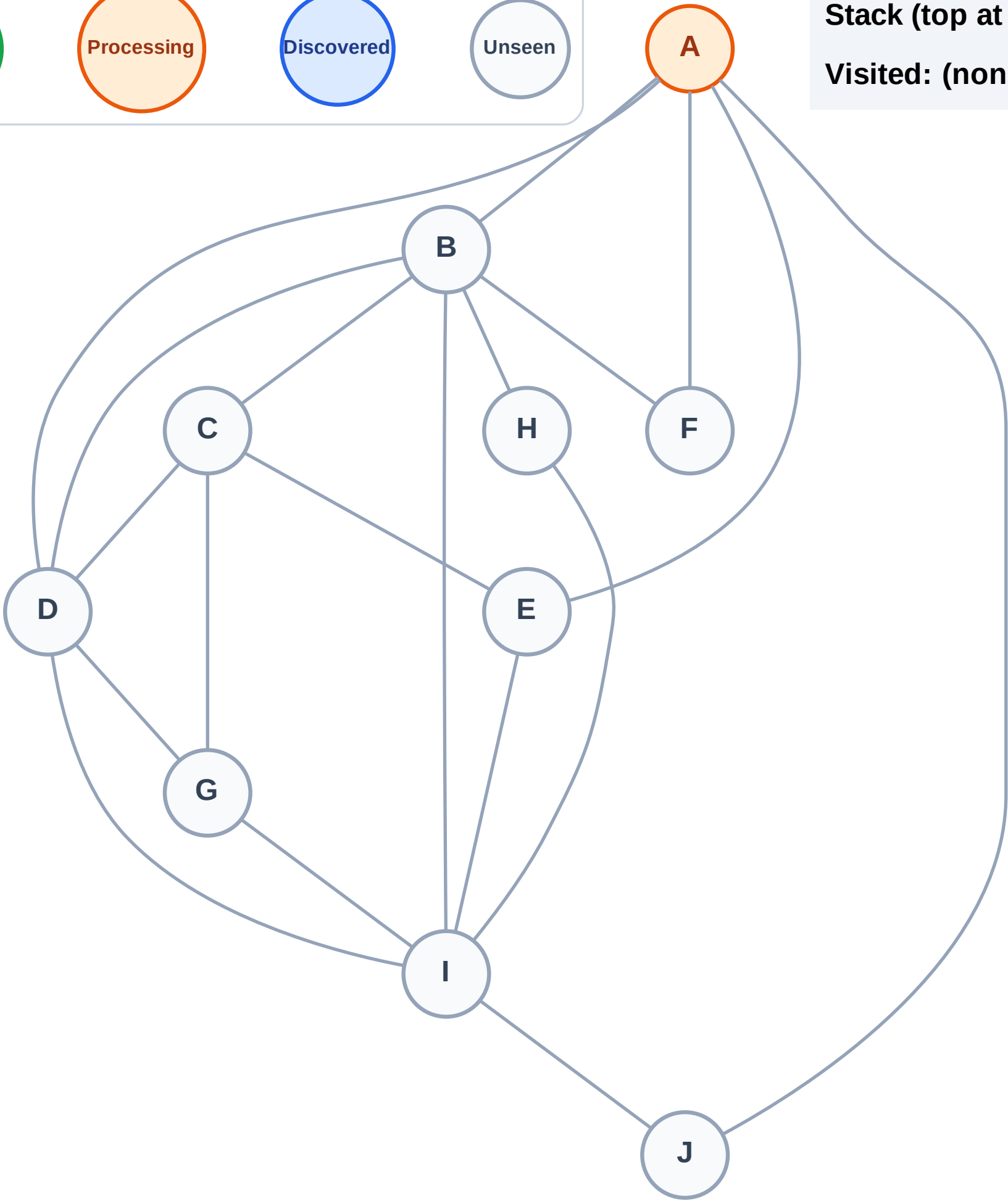
Complete

Processing

Discovered

Unseen

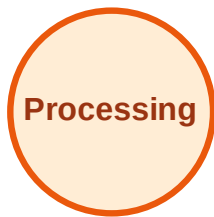
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

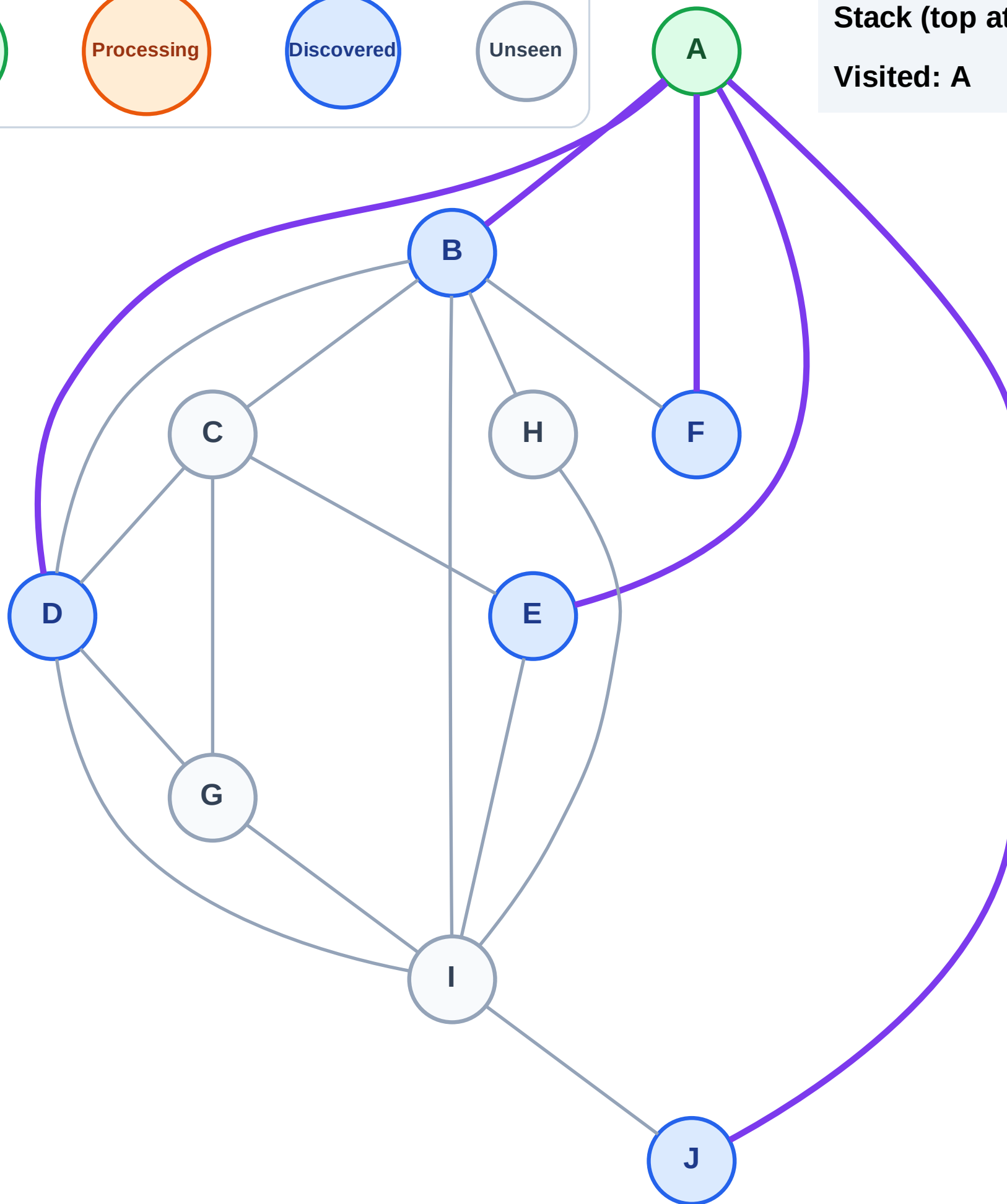
Step 3: Discover J, F, E, D, B from A

Legend



Stack (top at right): J | F | E | D | B

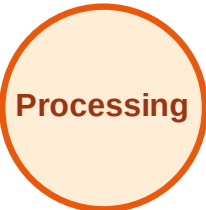
Visited: A



DFS Traversal

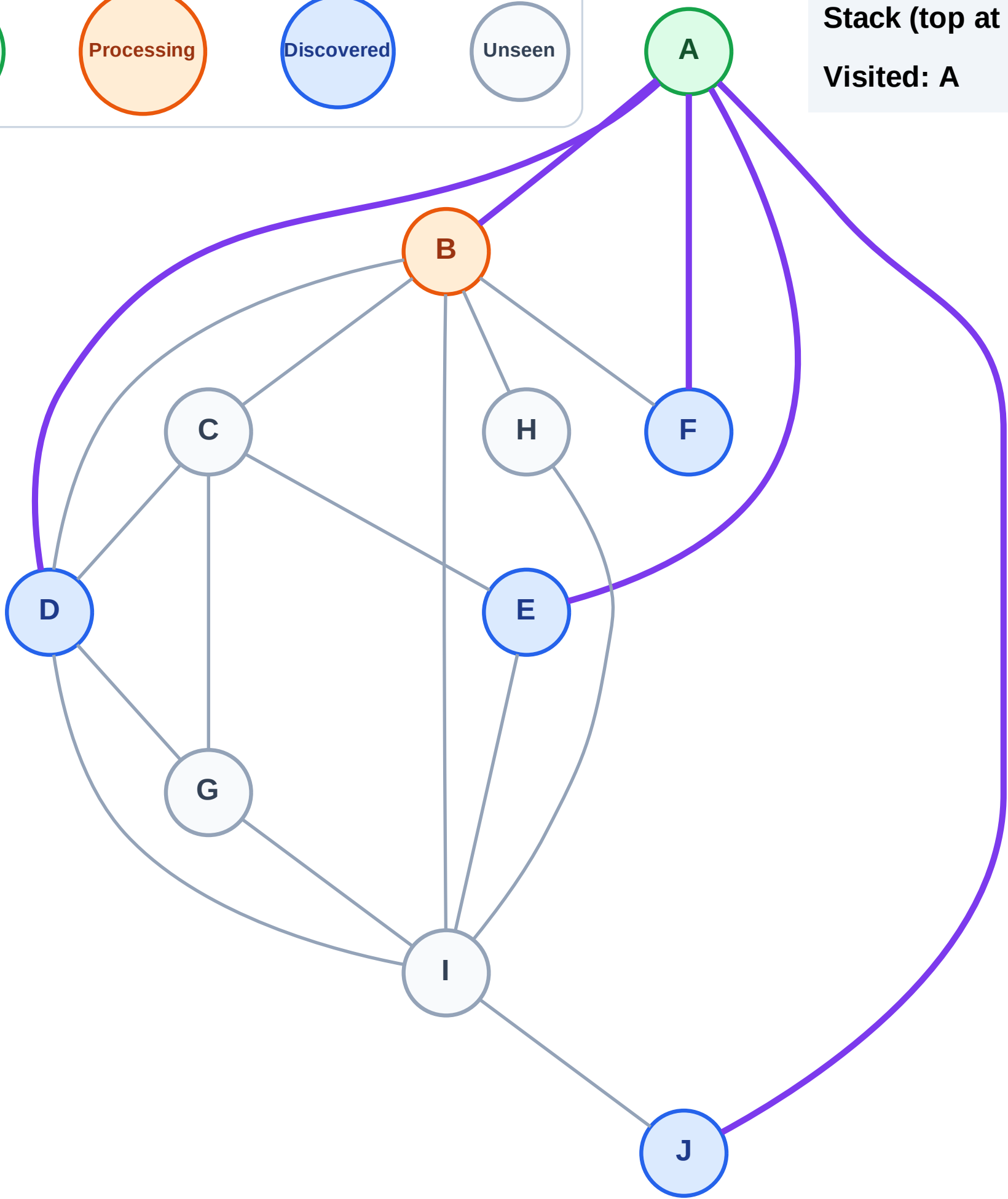
Step 4: Process node B

Legend



Stack (top at right): J | F | E | D

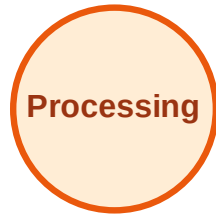
Visited: A



DFS Traversal

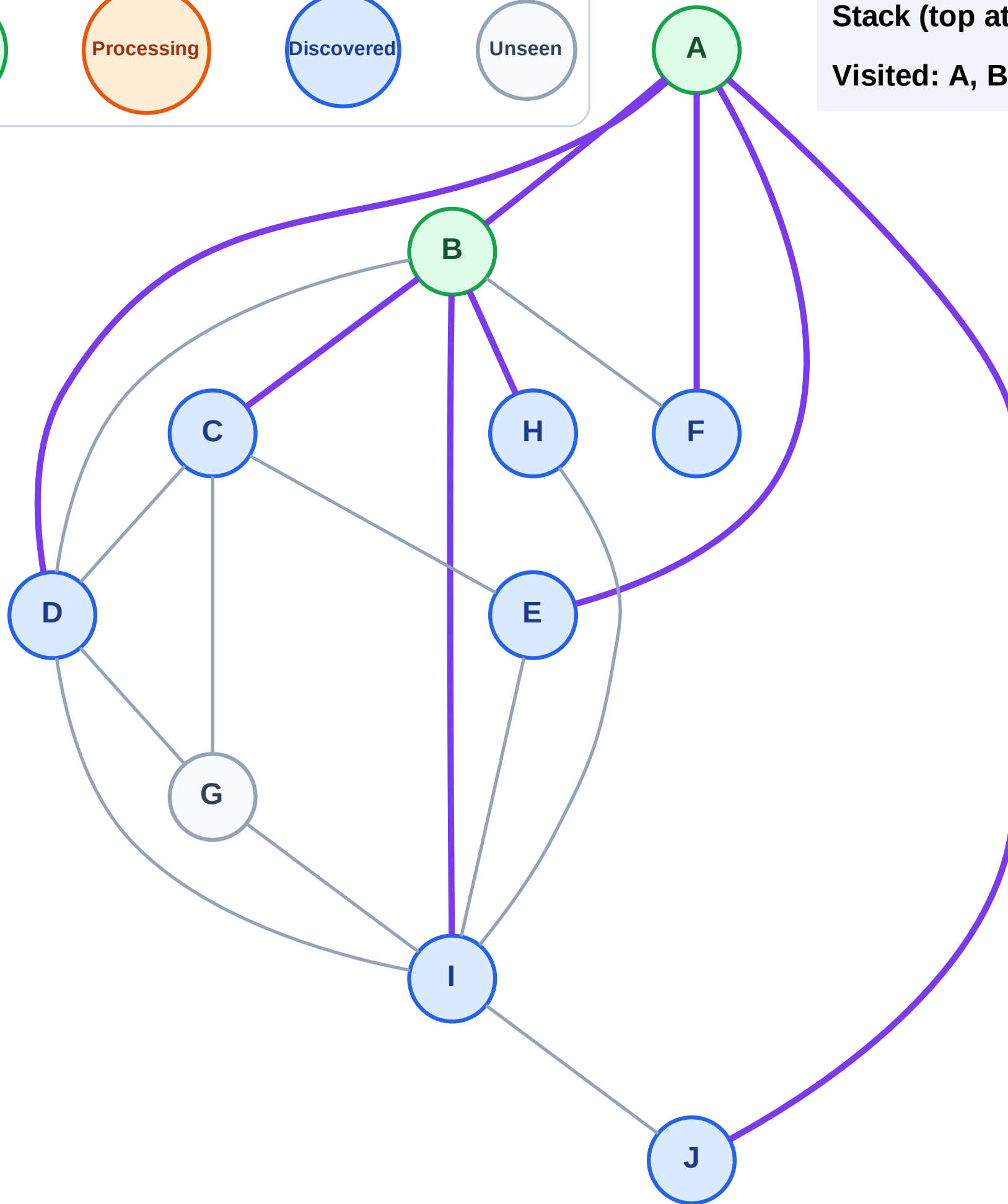
Step 5: Discover I, H, C from B

Legend



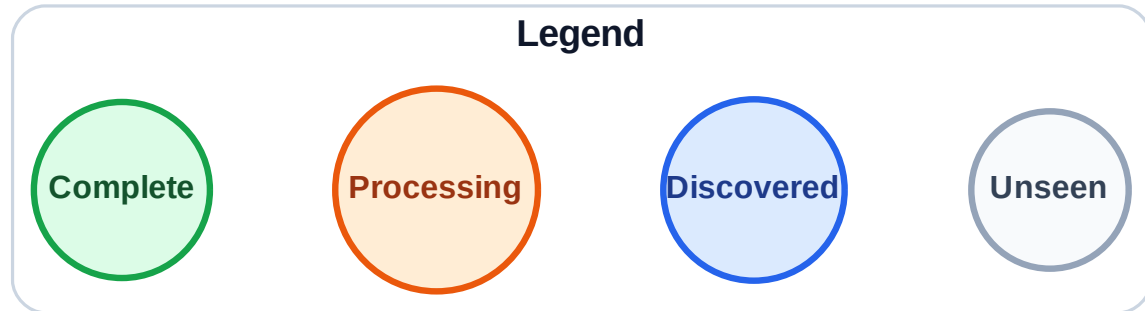
Stack (top at right): J | F | E | D | I | H | C

Visited: A, B



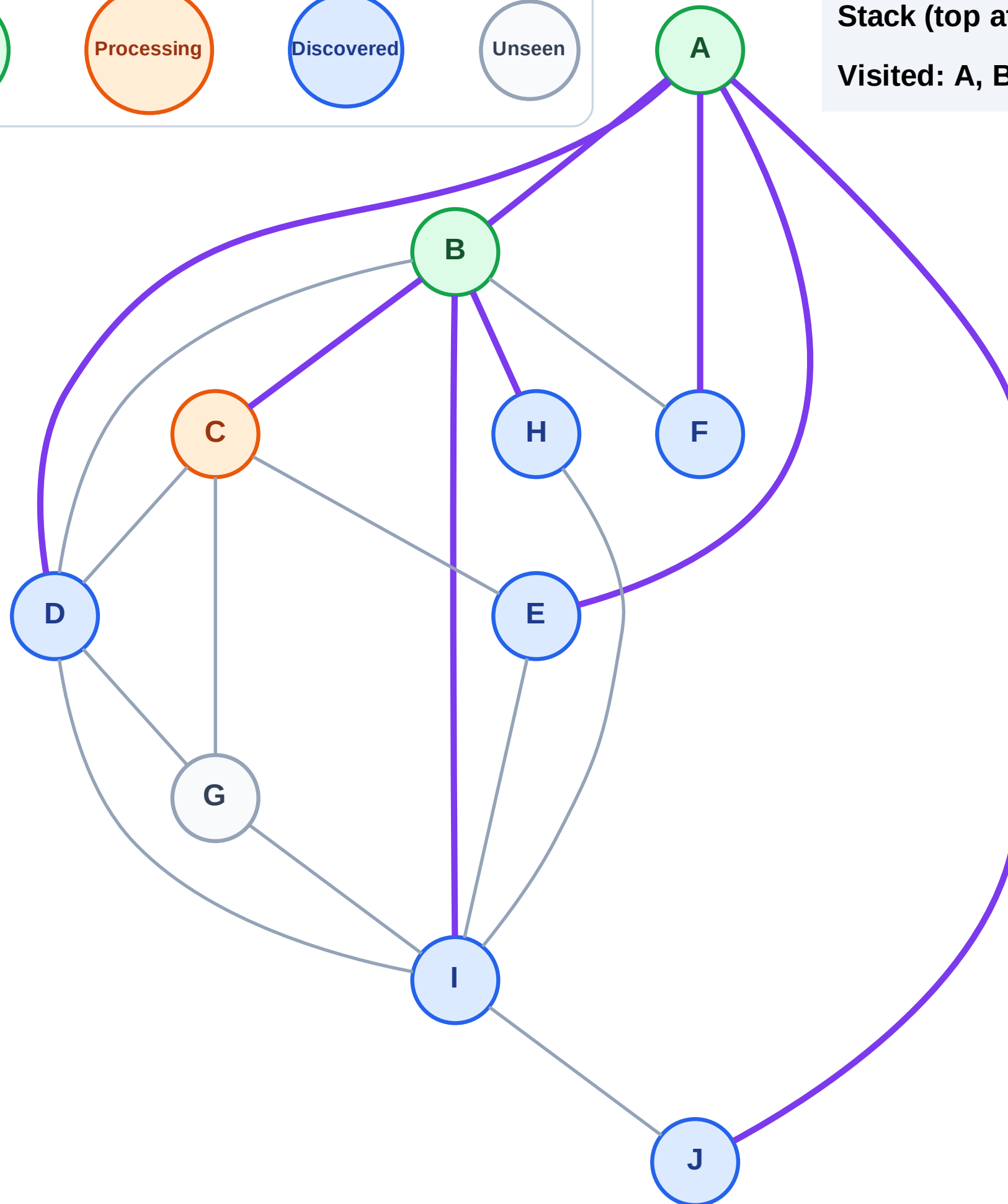
DFS Traversal

Step 6: Process node C



Stack (top at right): J | F | E | D | I | H

Visited: A, B



DFS Traversal

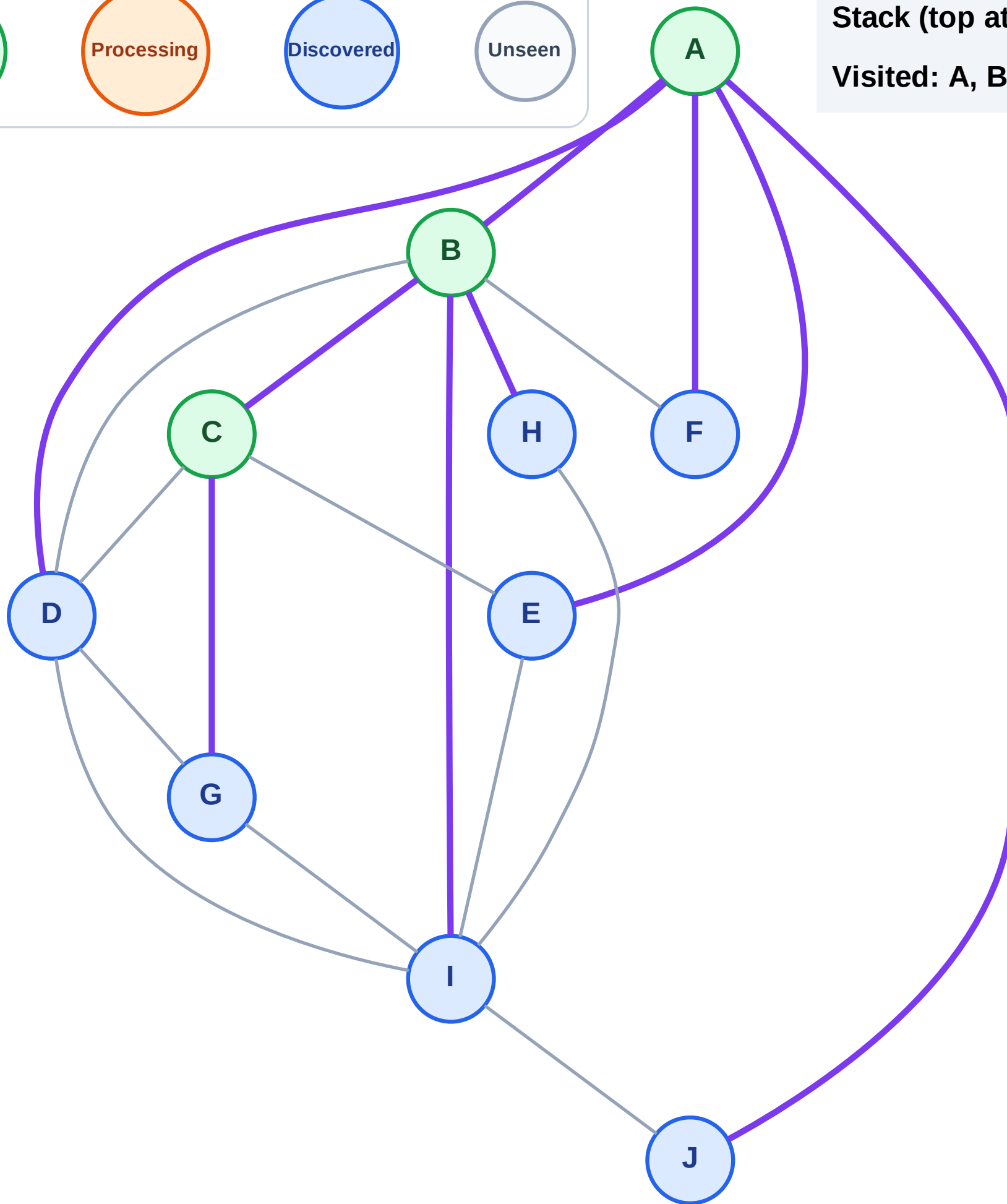
Step 7: Discover G from C

Legend



Stack (top at right): J | F | E | D | I | H | G

Visited: A, B, C



DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



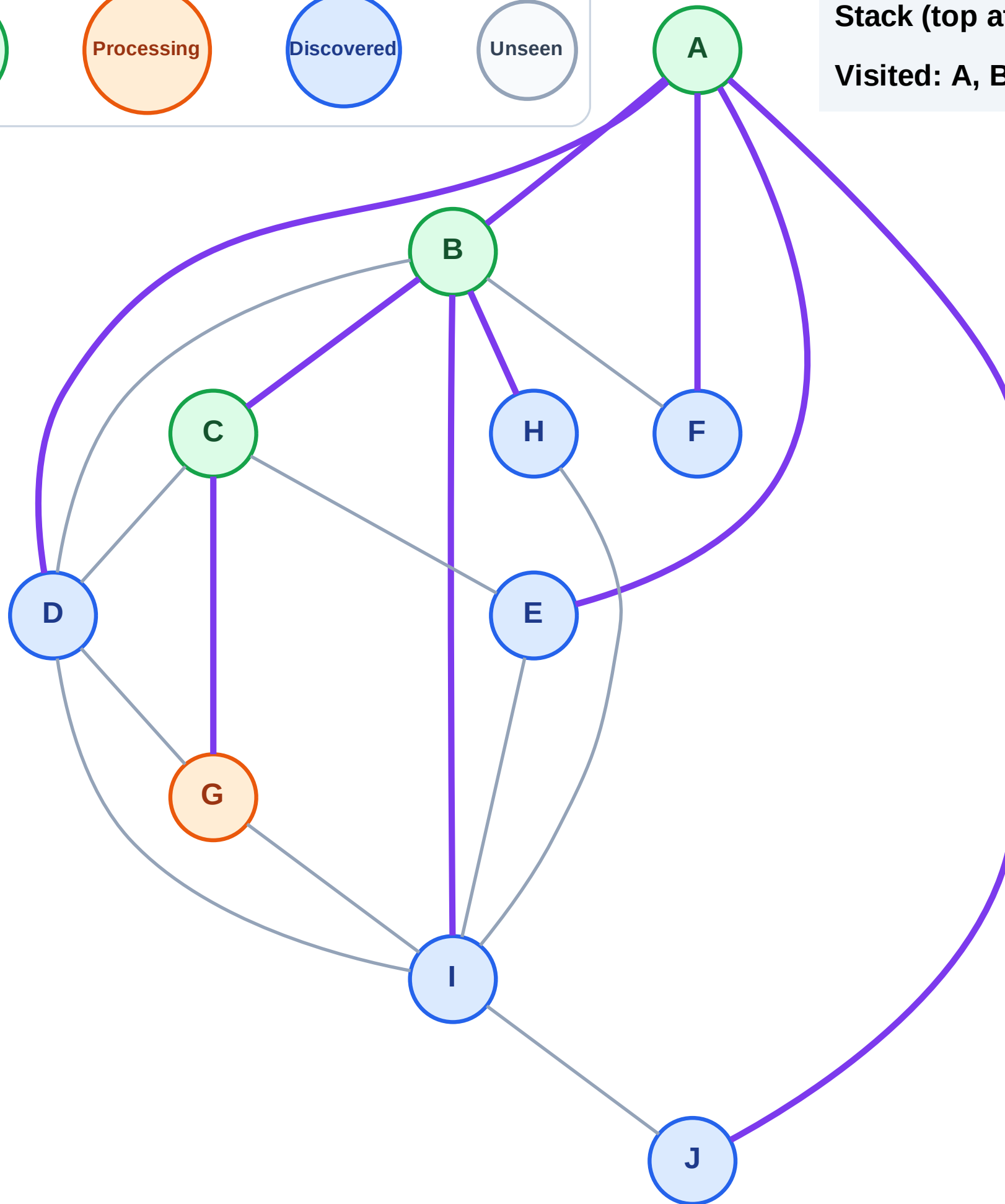
Discovered



Unseen

Stack (top at right): J | F | E | D | I | H

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from G

Legend



Complete



Processing



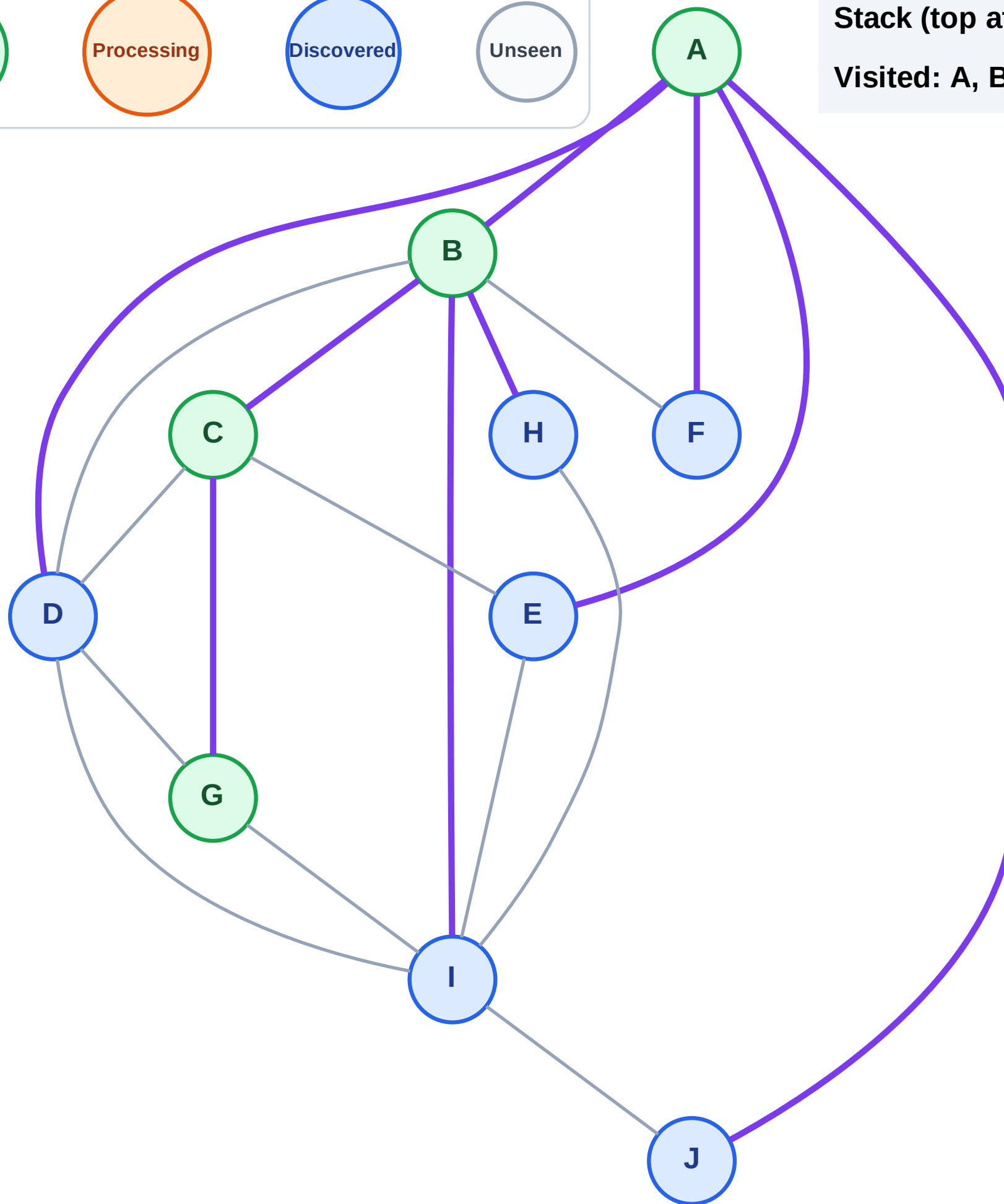
Discovered



Unseen

Stack (top at right): J | F | E | D | I | H

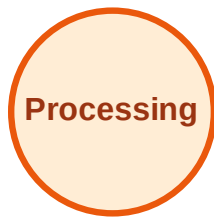
Visited: A, B, C, G



DFS Traversal

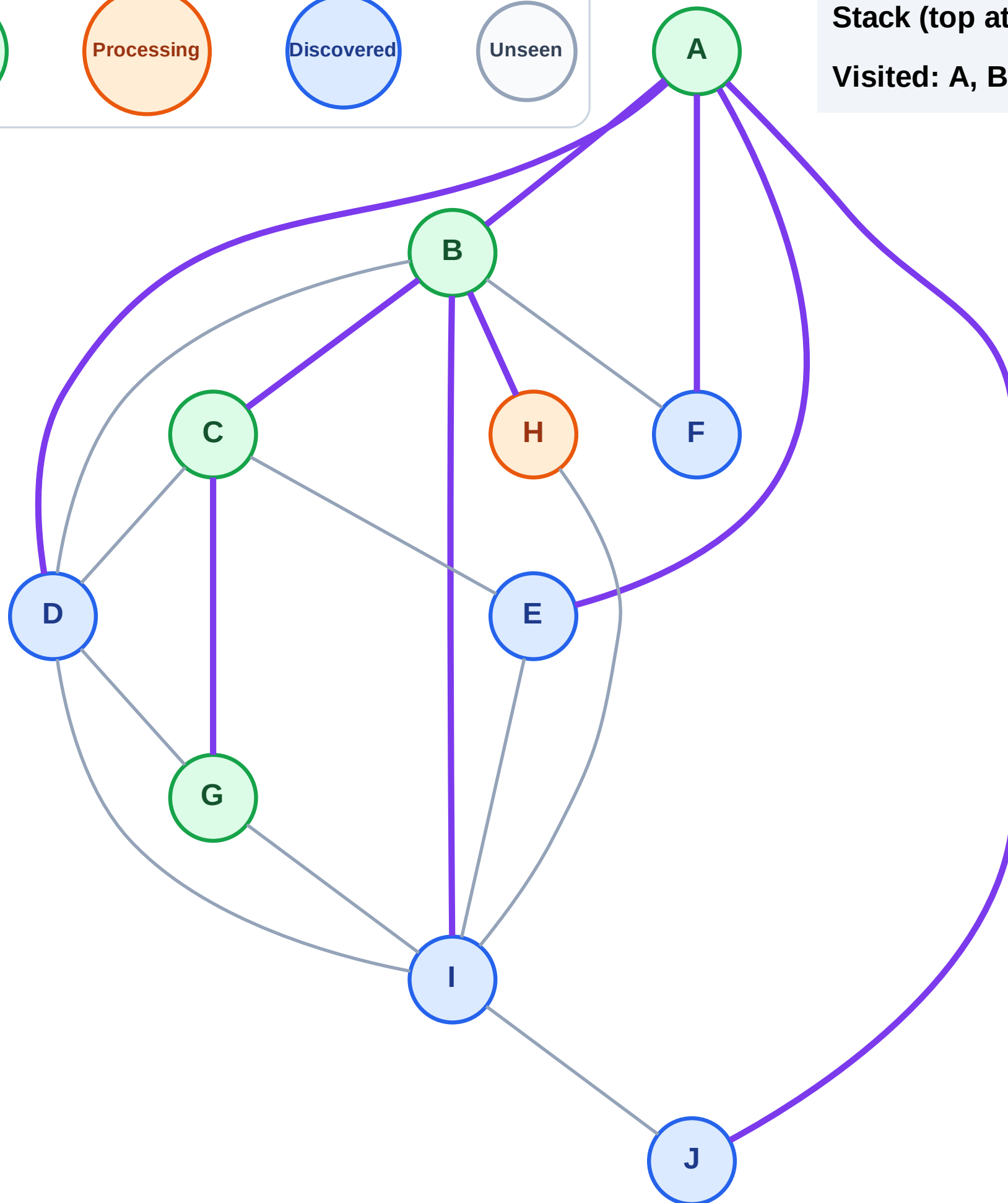
Step 10: Process node H

Legend



Stack (top at right): J | F | E | D | I

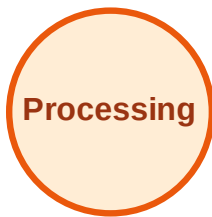
Visited: A, B, C, G



DFS Traversal

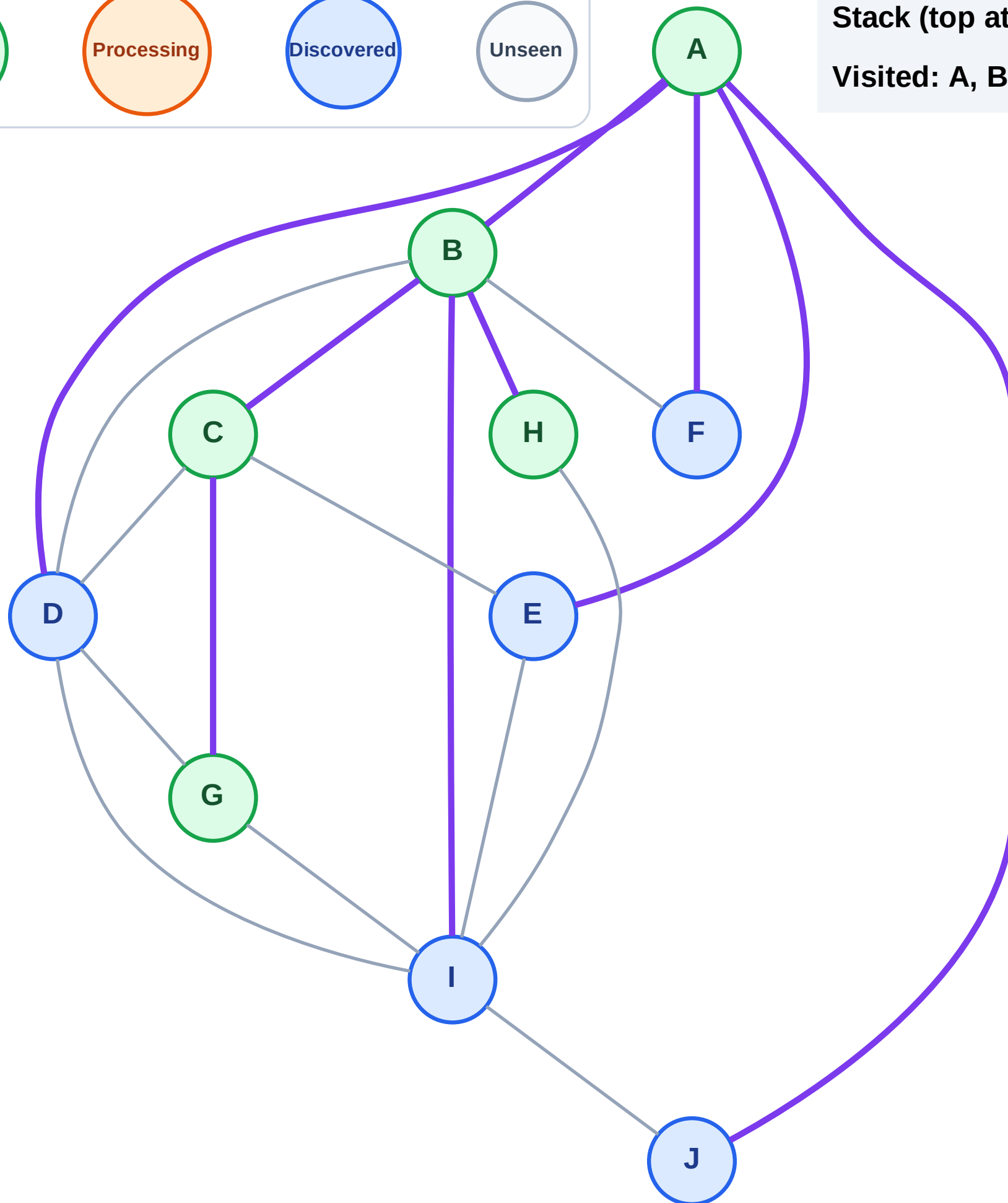
Step 11: No new neighbors from H

Legend



Stack (top at right): J | F | E | D | I

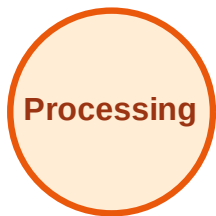
Visited: A, B, C, G, H



DFS Traversal

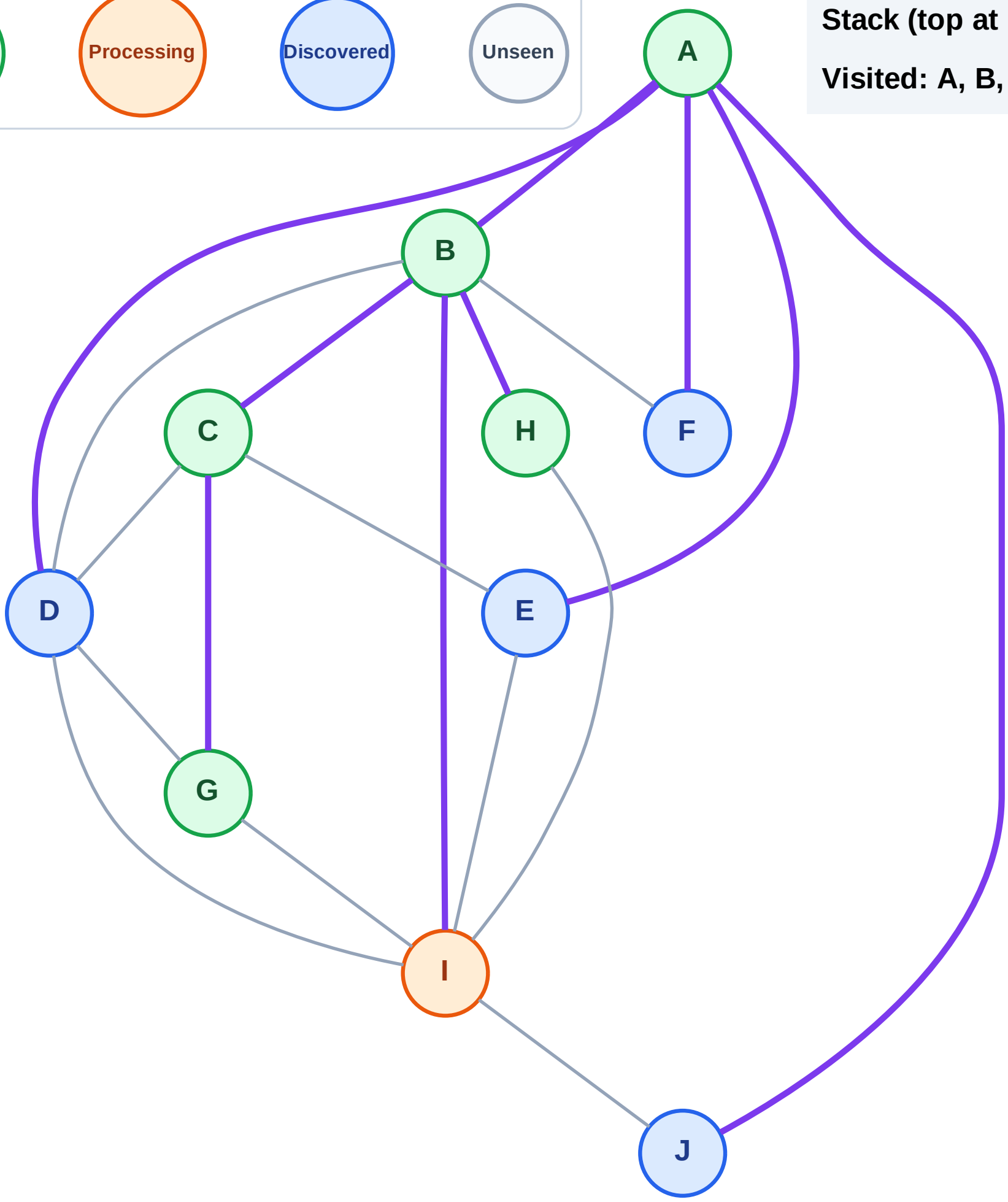
Step 12: Process node I

Legend



Stack (top at right): J | F | E | D

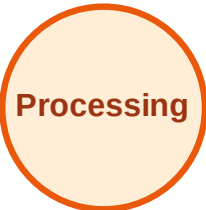
Visited: A, B, C, G, H



DFS Traversal

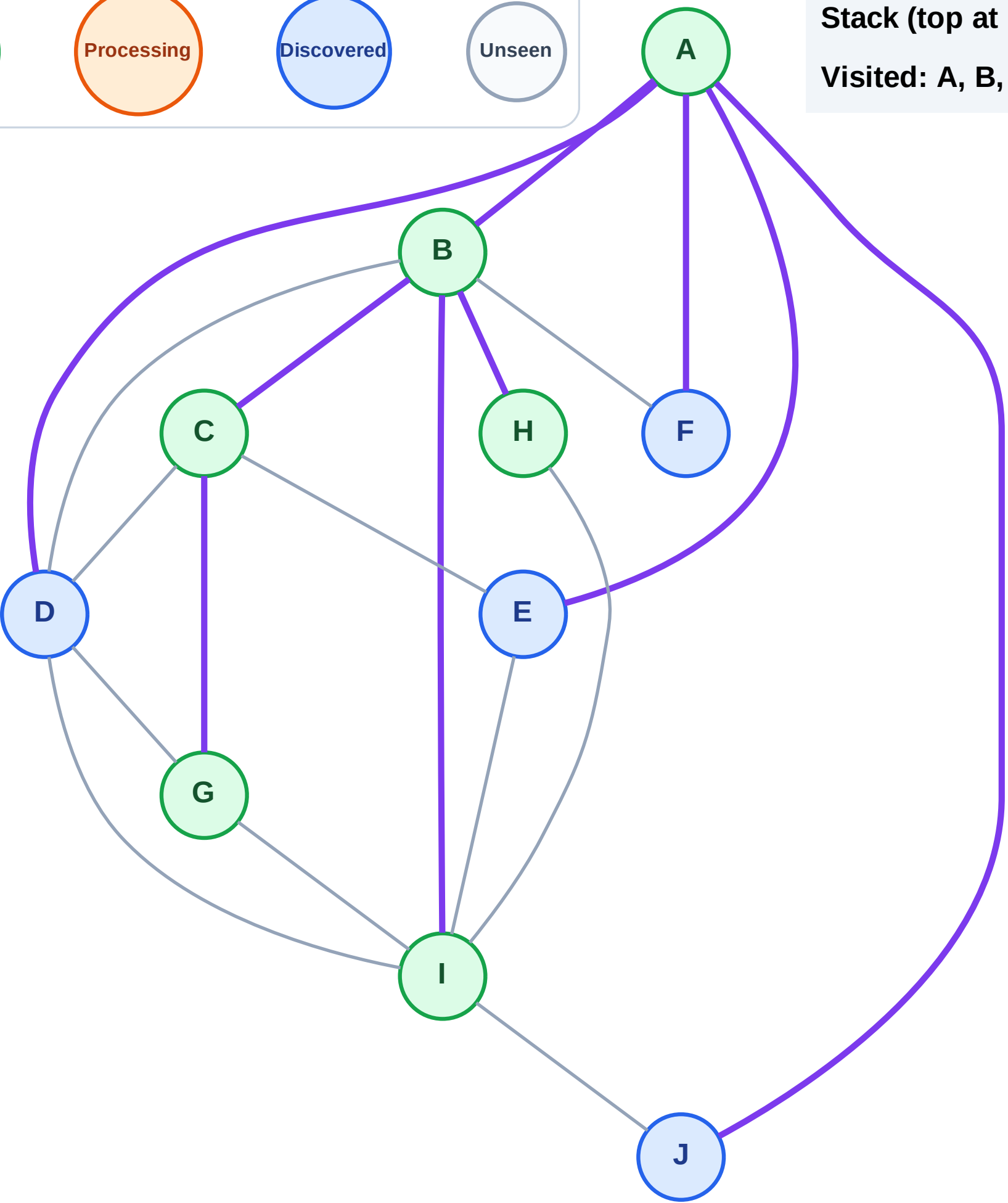
Step 13: No new neighbors from I

Legend



Stack (top at right): J | F | E | D

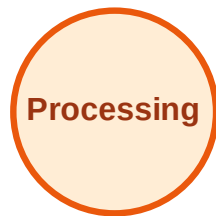
Visited: A, B, C, G, H, I



DFS Traversal

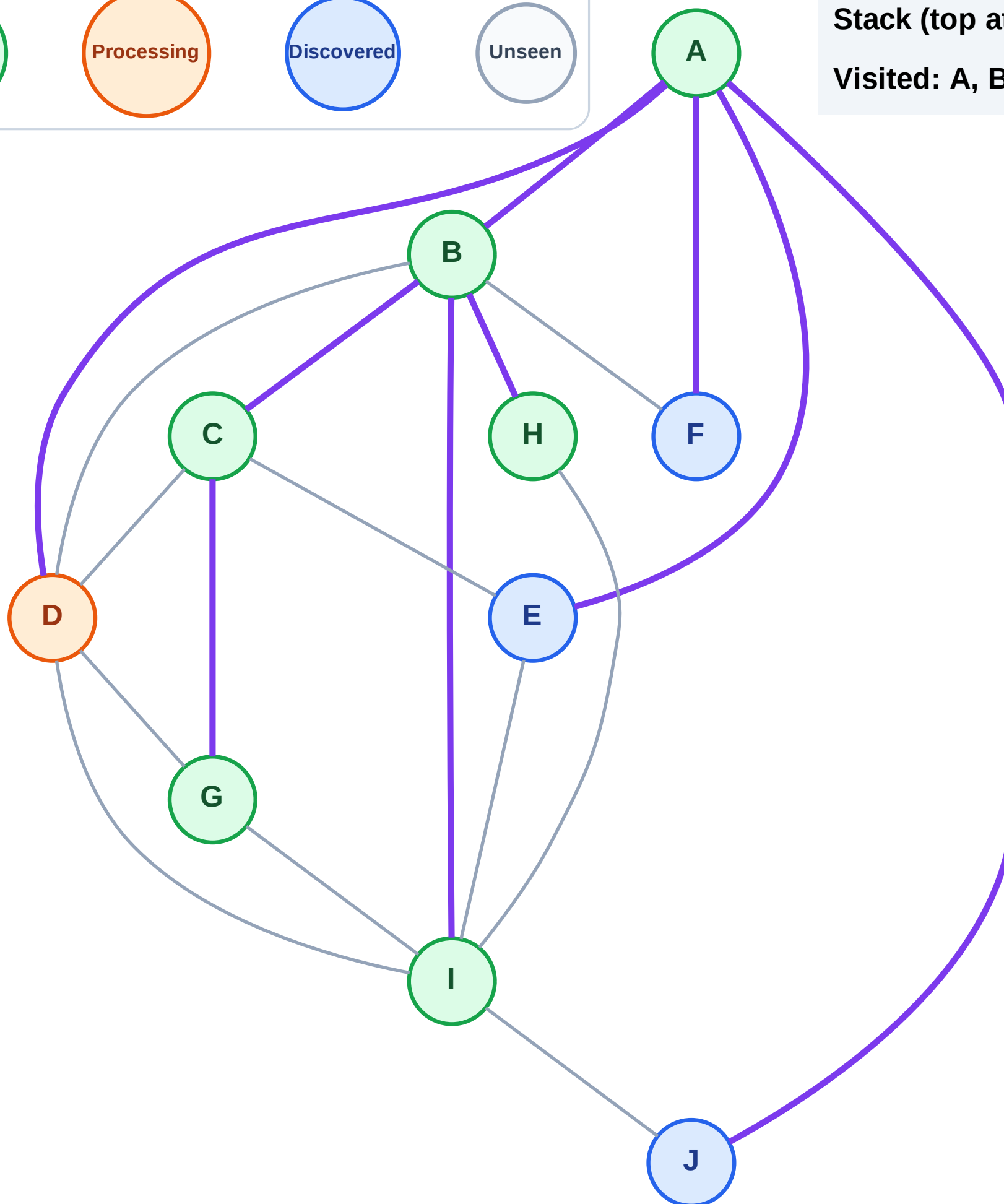
Step 14: Process node D

Legend



Stack (top at right): J | F | E

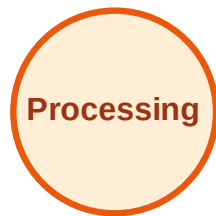
Visited: A, B, C, G, H, I



DFS Traversal

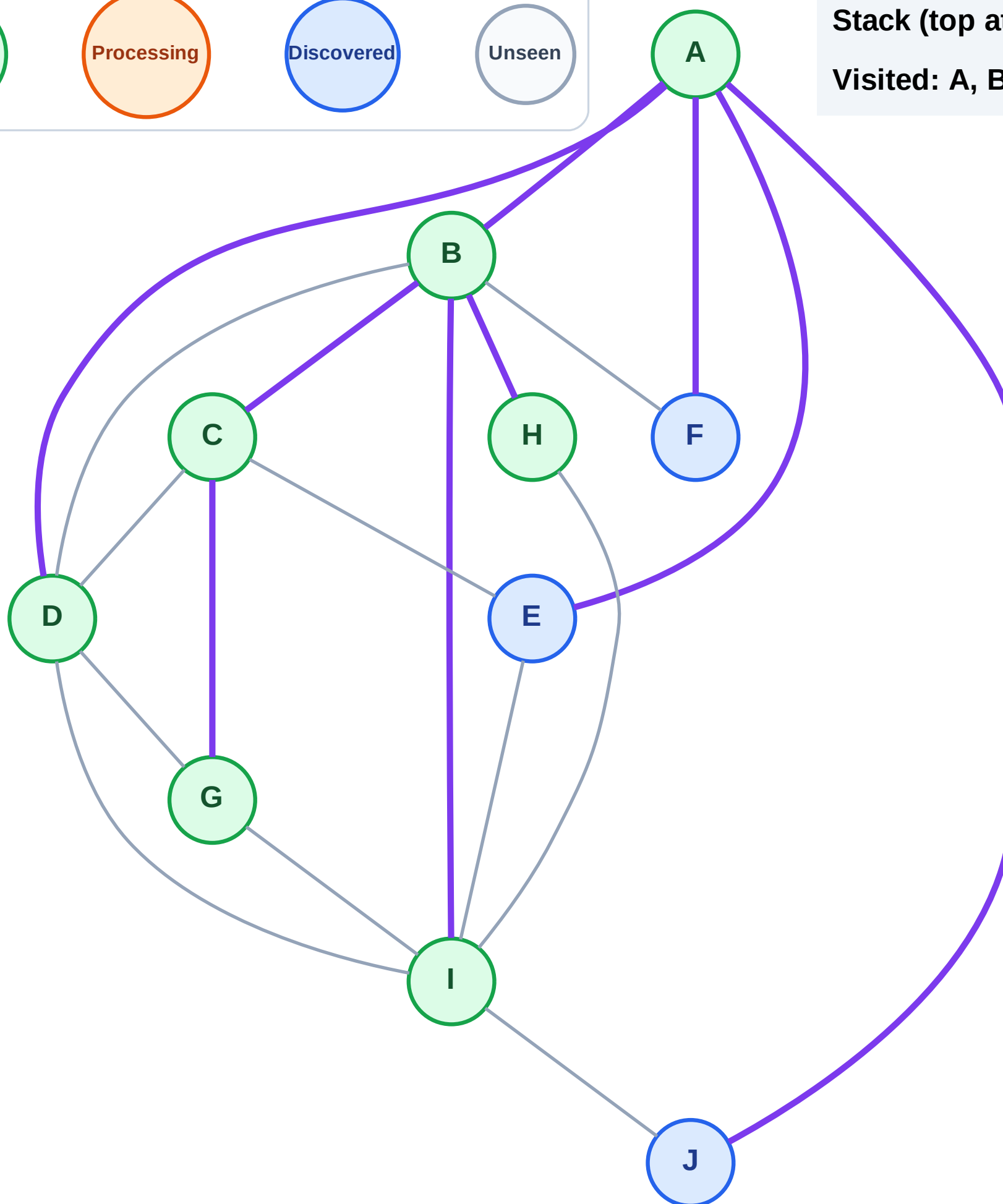
Step 15: No new neighbors from D

Legend



Stack (top at right): J | F | E

Visited: A, B, C, D, G, H, I



DFS Traversal

Step 16: Process node E

Legend

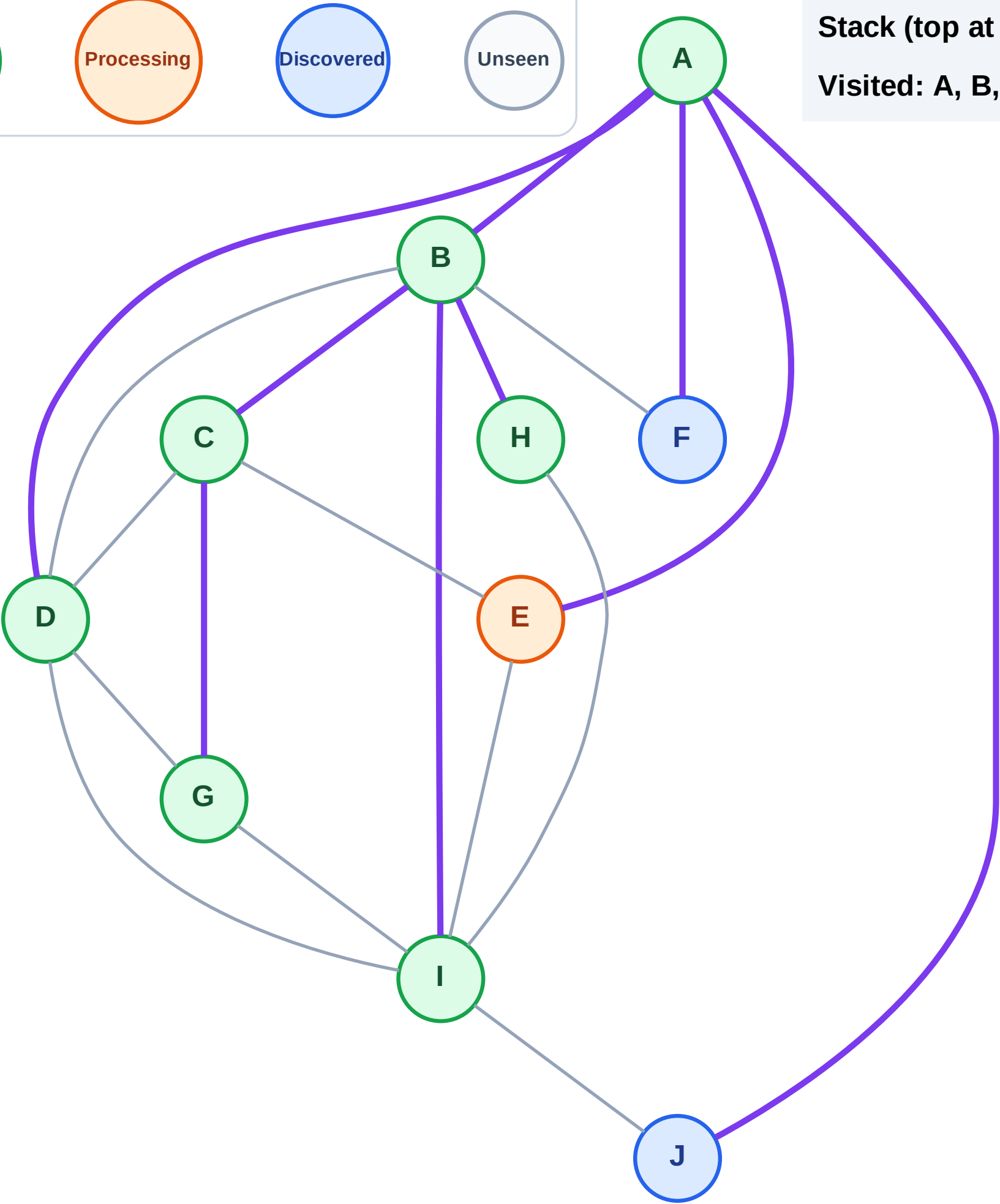
Complete

Processing

Discovered

Unseen

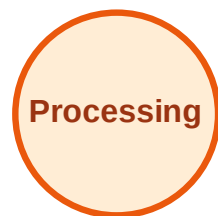
Stack (top at right): J | F
Visited: A, B, C, D, G, H, I



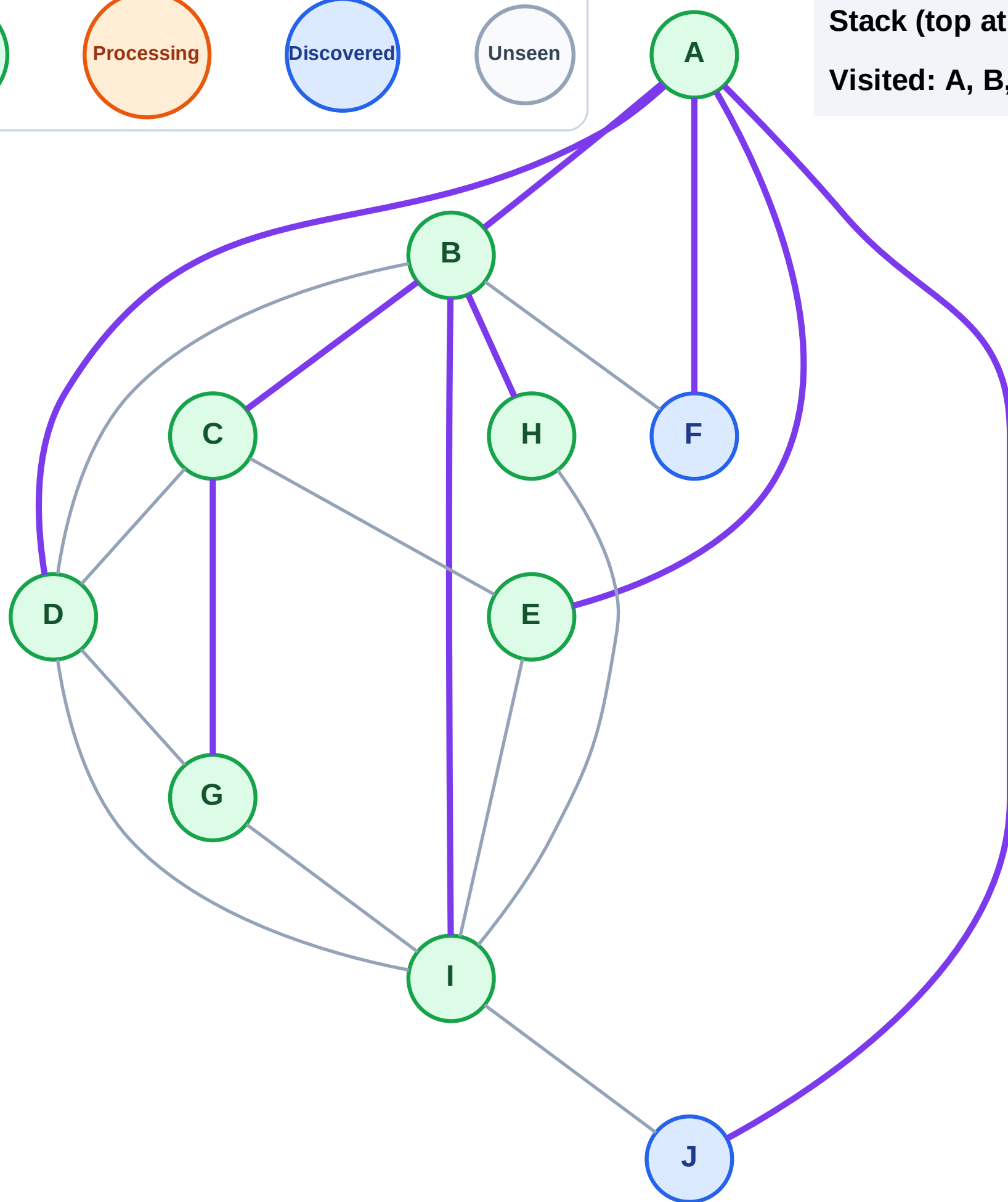
DFS Traversal

Step 17: No new neighbors from E

Legend



Stack (top at right): J | F
Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 18: Process node F

Legend

Complete

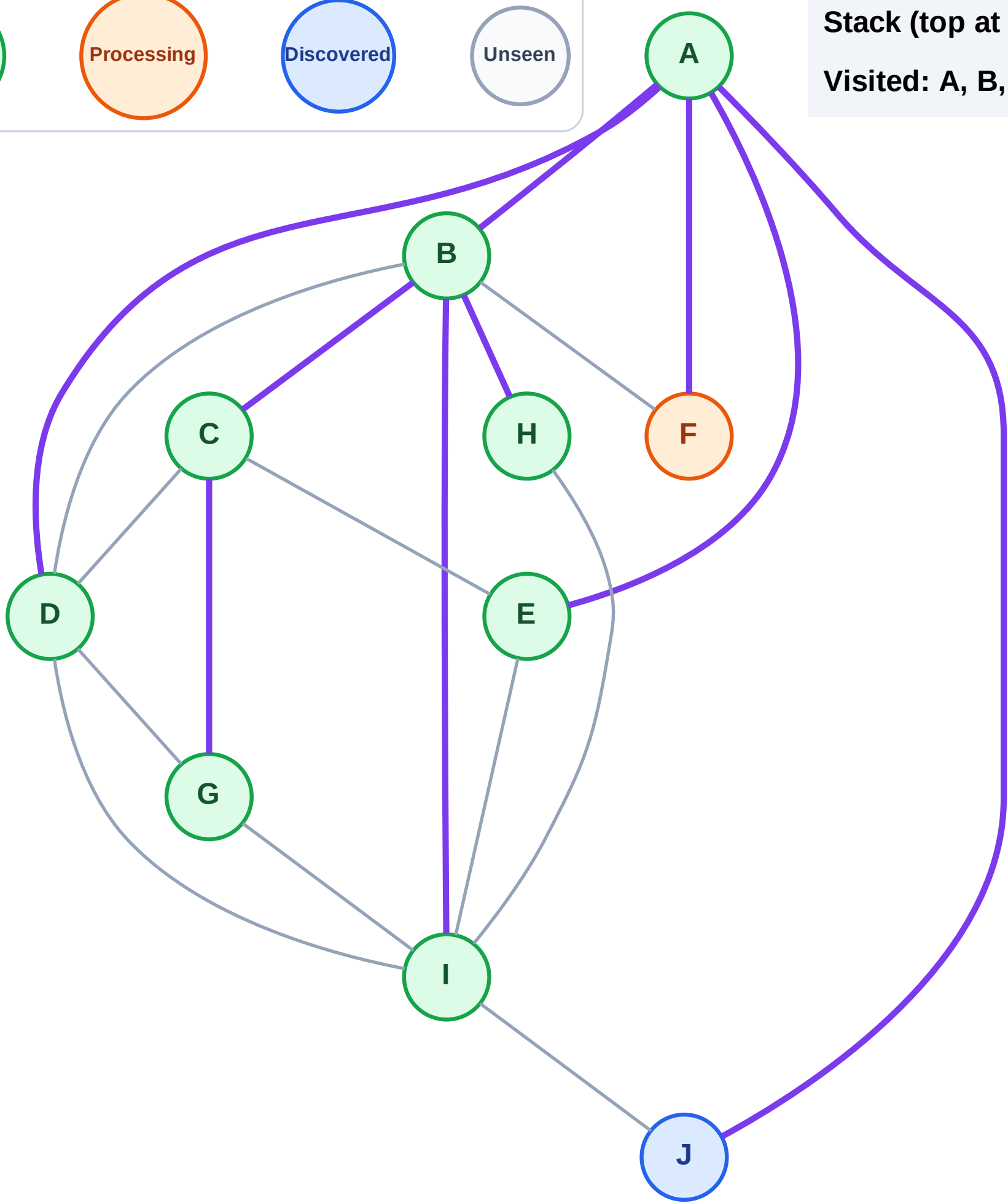
Processing

Discovered

Unseen

Stack (top at right): J

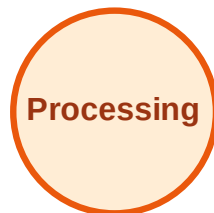
Visited: A, B, C, D, E, G, H, I



DFS Traversal

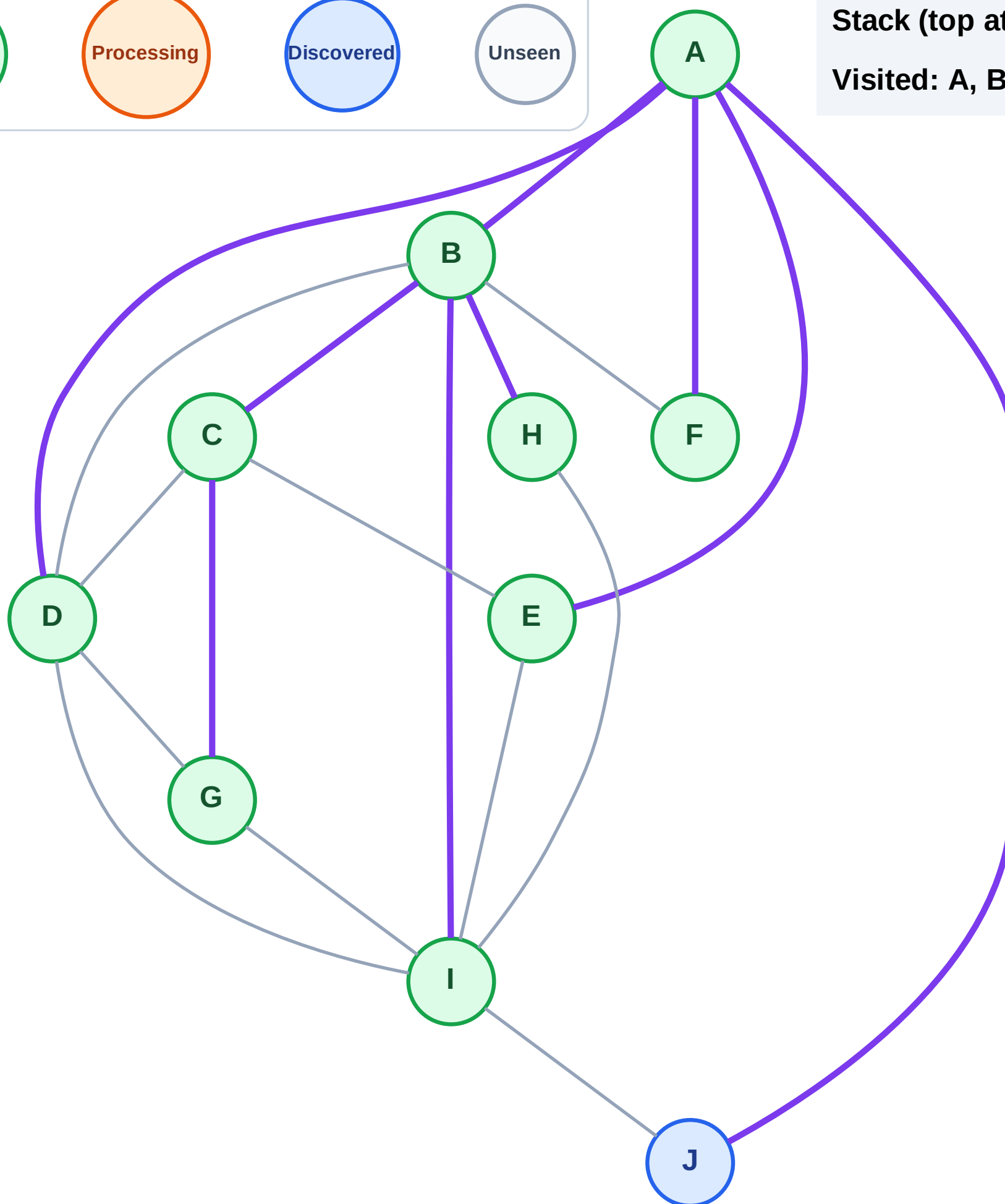
Step 19: No new neighbors from F

Legend



Stack (top at right): J

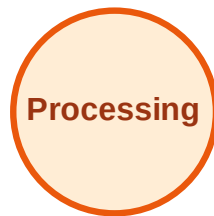
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

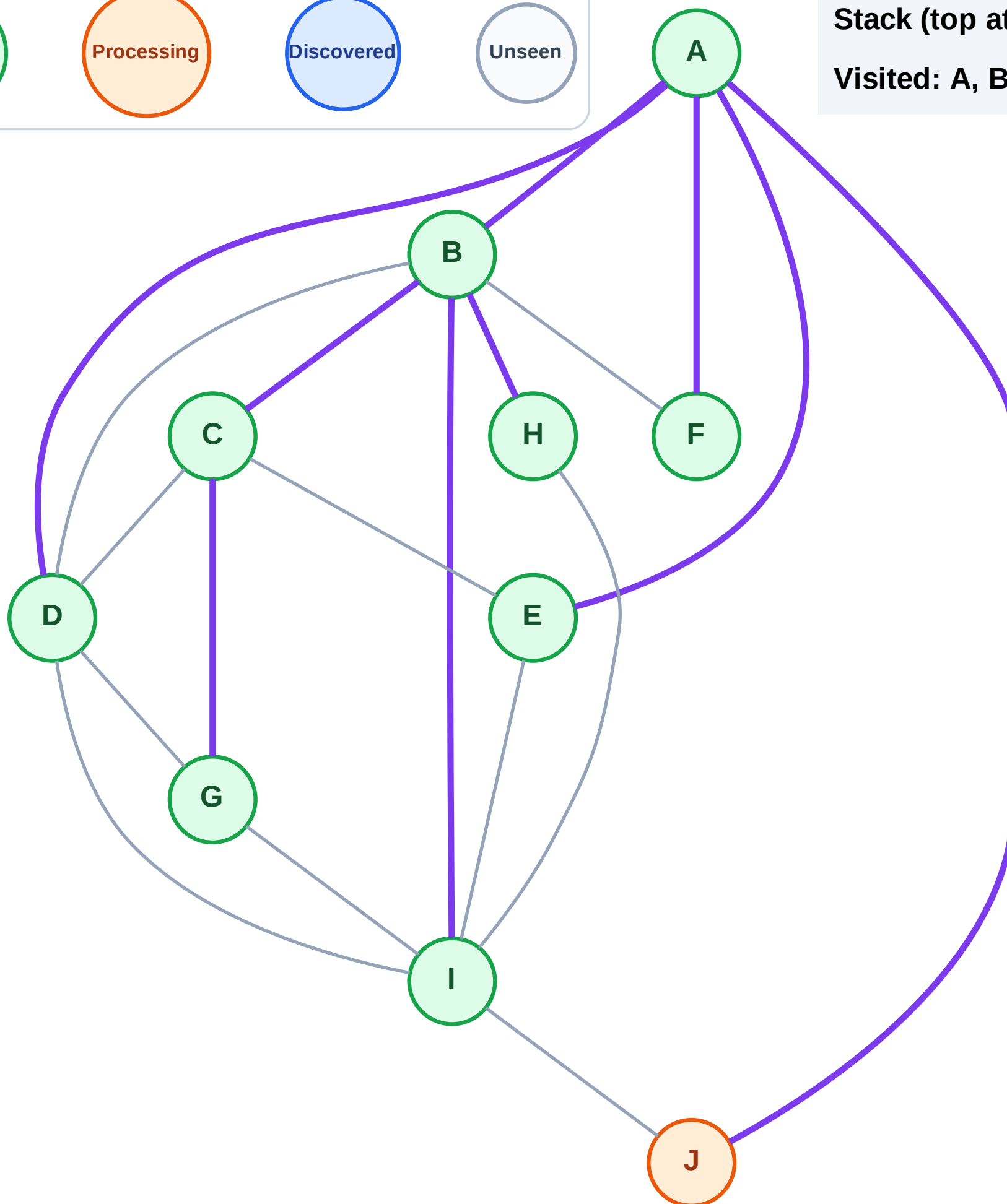
Step 20: Process node J

Legend



Stack (top at right): (empty)

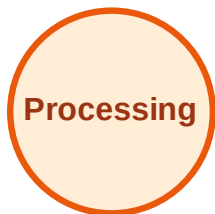
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

